



## Project: The Peaucellier-Lipkin Linkage

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## Abstract

This report presents the design, fabrication, and comprehensive analysis of a marble-lifting mechanism driven by the Peaucellier–Lipkin linkage, the first planar mechanism capable of generating exact straight-line motion from a rotary input. The project aims to demonstrate the practical implementation of this classical eight-bar linkage as a low-friction vertical elevator, highlighting its advantages over conventional slider-based mechanisms.

A physical prototype was constructed using an MDF base, acrylic links, 3D-printed components, and steel fasteners. Detailed CAD modeling was performed using SOLIDWORKS to generate orthographic views, assemblies, exploded views, and motion studies. The kinematic behavior of the mechanism was analytically investigated using complex vector-loop equations to determine the position, velocity, acceleration, and force distributions of all links. Numerical verification was conducted through MATLAB simulations, confirming the straight-line trajectory of the output point and providing corresponding velocity and acceleration profiles.

The results obtained from the physical prototype, CAD simulations, and mathematical analysis showed strong agreement, validating the theoretical performance of the Peaucellier–Lipkin linkage within acceptable experimental tolerances. Overall, the project successfully demonstrates the effectiveness of pure linkage-based solutions for precision motion generation with reduced friction and improved accuracy.

**Keywords:** Peaucellier–Lipkin linkage, straight-line mechanism, kinematic analysis, SOLIDWORKS, MATLAB simulation.

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# 1 Introduction

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## 1.1 Background

A central challenge in the history of machine design has been converting rotary motion easily generated by cranks and motors into straight-line motion. While the slider-crank mechanism is the most common solution, it introduces significant side thrust and friction due to the sliding contact, not to mention the need for an already established straight surface or a constraint to carry out the motion on [1].

To mitigate these issues, engineers sought pure linkage solutions. The Peaucellier-Lipkin mechanism (commonly referred to as the Peaucellier mechanism), an eight-bar linkage invented independently by Charles-Nicolas Peaucellier in 1864 and Lipkin in 1871, was the first planar linkage capable of converting circular motion into a perfectly straight line without the need of external constraints [2].

Unlike the earlier Watt Linkage, which only produced an approximate straight line, the Peaucellier-Lipkin linkage produces a true, geometrically sound, exact straight line. This project utilizes this classical mechanism as the driving force of a marble machine, which can be translated into various applications.

## 1.2 Practical Importance

Understanding straight-line mechanisms is critical for designing machinery where precision and low friction are necessary.

In industrial applications, exact straight-line mechanisms can eliminate the need for sliders, hence reducing wear, lubrication requirements, and manufacturing costs associated with high-precision linear guides (constraints) [3].

In this specific project, the Peaucellier mechanism functions as a **vertical lift elevator**, and other applications (e.g, **mechanical drawing and drafting instruments**, **precision motion demonstration devices** and **robotics and mechanism research**) . The linkage utilizes its exact straight-line output to raise a platform carrying marbles to the intake of a metal track. This application demonstrates the mechanism's ability to maintain verticality without lateral deviation, ensuring a smooth transfer of the marbles without the friction of vertical guide rails if a crank-slider linkage were used.

### 1.3 Purpose of the Report

The primary objective of this report is to document the complete design lifecycle of this Peaucellier-based marble machine. This includes:

- i. The theoretical kinematic and kinetic analysis
- ii. Computer-aided design ”CAD”
- iii. Numerical simulation and motion verification using **MATLAB** and the physical construction of the working model.

The report aims to verify that the physical prototype adheres to the theoretical results produced by the software within acceptable standards.

### 1.4 Report Scope and Structure

This report is organized to guide the reader through the design process sequentially:

- i. **The Design:** Details the physical realization of the model, including material selection and fabrication techniques, alongside the SOLIDWORKS three views, assembly, exploded views, plus motion analysis.
- ii. **Kinematics:** Provides the mathematical foundation, analyzing the position, velocity, acceleration, and motion of the mechanism to verify straight-line generation, along with the kinetic analysis to ensure the prototype is structurally sound under the acting forces of movement and inertia, along with the weight of the transferred masses.
- iii. **MATLAB Simulation:** Presents the code used to plot trajectory and velocity profiles, serving as a verification tool for the theoretical analysis, and an ideal-case demonstration of the mechanism.
- iv. **Conclusion:** Summarizes the results, discussing the accuracy of the motion and the correlation between the physical model, CAD data, and mathematical simulations.
- v. **References:** Lists all academic sources used throughout the report for further consultation.

## 2 The Design

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### 2.1 Physical Model

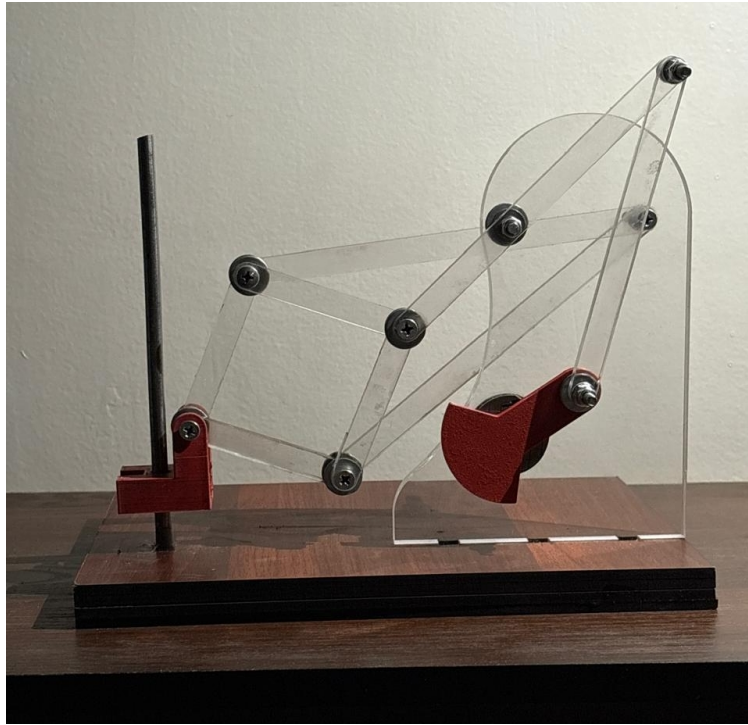


Figure 1: The Peaucellier-Lipkin Linkage Project

#### 2.1.1 Material Selection

Material selection for the Peaucellier–Lipkin linkage was carried out with consideration of mechanical strength, manufacturability, cost, and suitability for an educational prototype. The selected materials ensured adequate rigidity and accuracy while remaining easy to fabricate within the available resources.

- i. **Base Plate:** Medium-Density Fiberboard (MDF) was used for the base of the mechanism due to its low cost, sufficient stiffness for laboratory-scale loads, and ease of fabrication. MDF is well-suited for laser cutting, which allowed accurate positioning of fixed joints and simplified the manufacturing process.
- ii. **Links:** Most of the links were manufactured from acrylic (PMMA). Acrylic provides sufficient strength and rigidity for the applied loads while maintaining a relatively low weight. Its smooth surface contributes to reduced friction at joints, improving the overall motion quality. Additionally, the transparency of acrylic (with light transmittance of up to 92%) allows clear visualization of the mechanism's motion, which is advantageous for demonstration and educational purposes.



- iii. **Fasteners (Nuts and Bolts):** Steel fasteners were used to assemble the links and joints due to their high strength and durability. In locations where loosening could occur due to repeated motion, self-locking counter nuts (pal nuts) were employed. These fasteners help maintain joint integrity and prevent loosening under dynamic conditions.
- iv. **Input Crank (OA) and Slider:** The input crank and slider were fabricated from polylactic acid (PLA) using 3D printing. This manufacturing method was selected due to its low cost, rapid prototyping capability, and suitability for producing geometrically complex components. The mechanical properties of PLA were sufficient for the relatively low loads applied to these components in the prototype.

### 2.1.2 Description of the linkage's motion

The assembled mechanism consists of two main subsystems: a four-bar crank–rocker mechanism and a Peaucellier–Lipkin linkage, both mounted on a common rigid base. The purpose of this combined assembly is to transform the continuous rotary motion of the input crank into an accurate straight-line motion at the output point.

The assembly process begins with the base plate, which serves as the fixed reference frame for the entire mechanism. All grounded components, including the motor support, fixed pivots, and the vertical guide of the slider, are rigidly attached to the base to ensure stability and proper alignment during operation.

The four-bar crank–rocker mechanism is assembled first. The input crank is connected directly to the motor shaft, allowing the motor's rotational motion to be transmitted to the mechanism. The crank is pinned to the coupler link using a revolute joint, while the coupler is connected to the rocker link, which oscillates about a fixed pivot mounted on the base. This subsystem converts the continuous rotation of the motor into a controlled oscillatory motion, which is required to drive the Peaucellier linkage smoothly.

The Peaucellier–Lipkin linkage is then assembled and attached to the rocker. Its links are connected using pin joints to form the characteristic rhombus and supporting arms that define the linkage geometry. Care is taken to ensure equal link lengths where required, as this geometric condition is essential for achieving the theoretical straight-line motion.

At the output side, the endpoint of the Peaucellier linkage is connected to a vertical slider, which is constrained to move only along a straight vertical guide. This guide eliminates any horizontal displacement and ensures that the generated motion closely approximates an ideal straight line. The slider is free to translate vertically while remaining perpendicular to the base throughout the motion cycle.

All joints are assembled using nuts and bolts, with self-locking (pal) nuts applied where vibration or repeated motion could cause loosening. This ensures reliable operation while

still allowing disassembly for maintenance or adjustment.

Once fully assembled, the mechanism operates as follows: the motor rotates the crank, the four-bar mechanism transmits and shapes the motion, and the Peaucellier linkage converts it into an accurate linear displacement at the slider. This integrated assembly demonstrates the practical implementation of classical kinematic theory using a combination of rotational and straight-line mechanisms.

## 2.2 SOLIDWORKS

### 2.2.1 Parts and Assembly

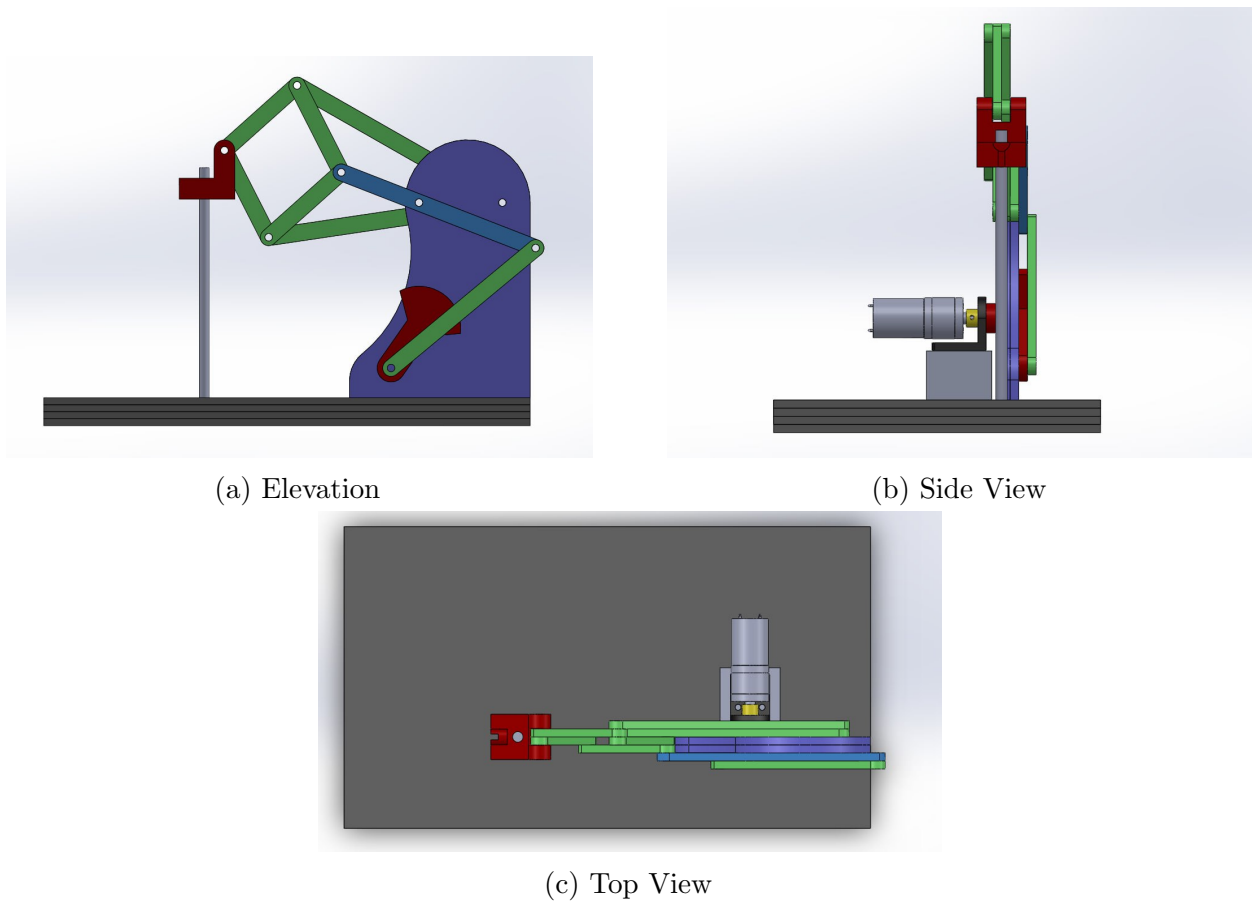
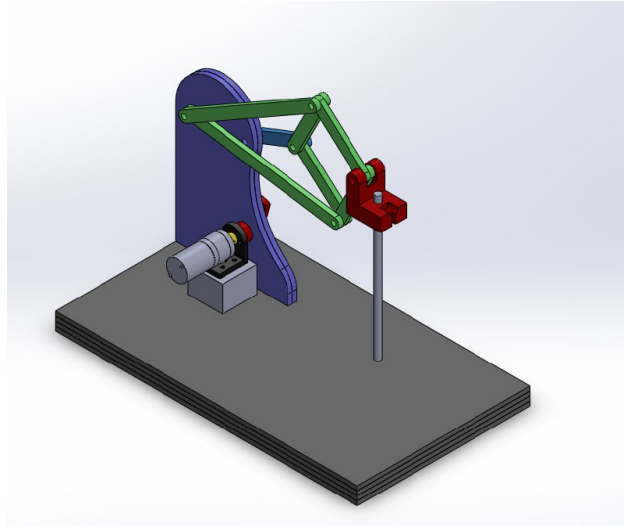
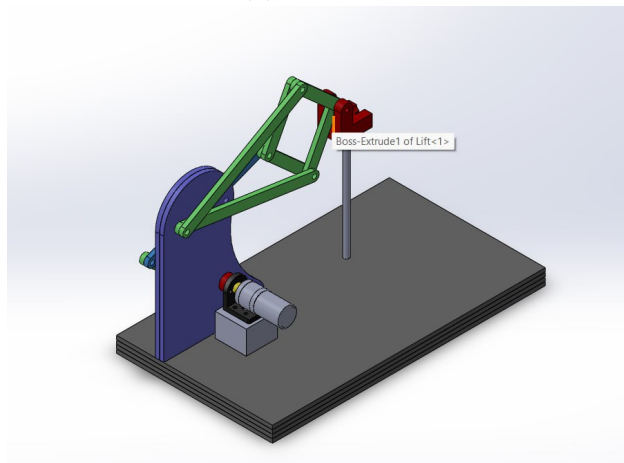


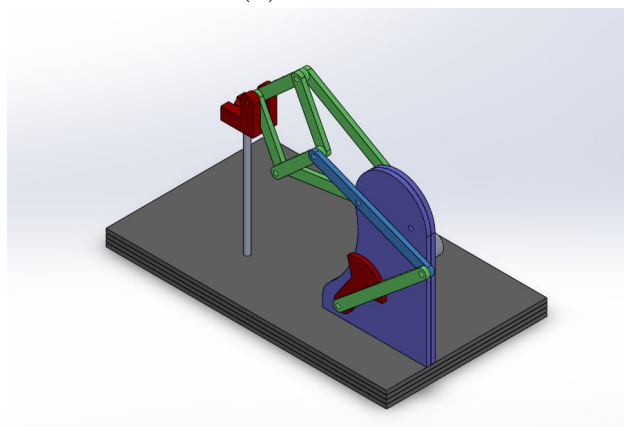
Figure 2: Comparison of the three views of the mechanism



(a) Assem 1



(b) Assem 2



(c) Assem 3

Figure 3: Different angles for the assembly of the linkage

### 2.2.2 Exploded Video

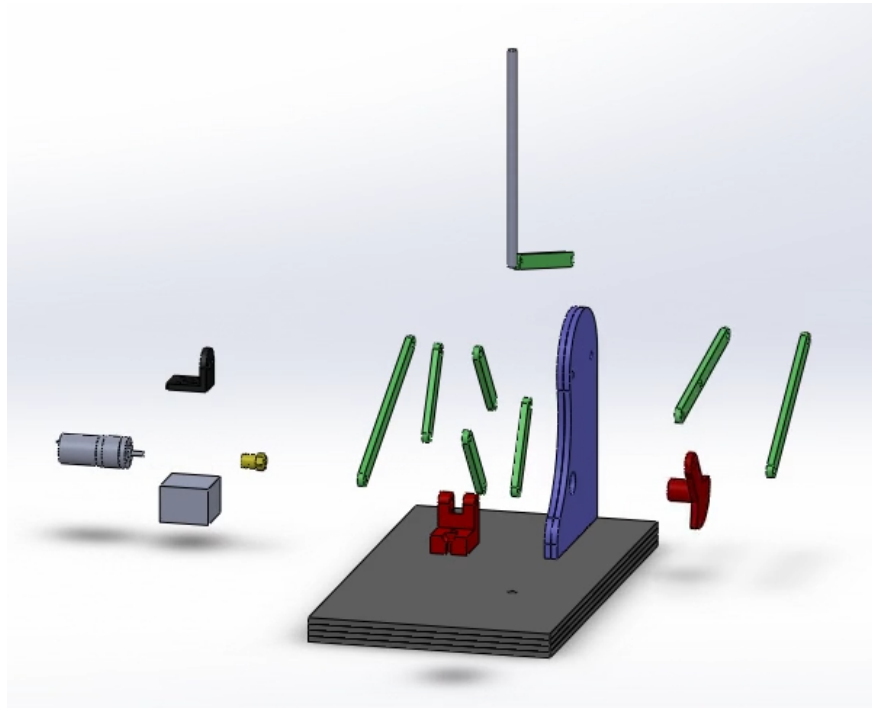


Figure 4: The exploded video for the linkage.

### 2.2.3 Motion Video

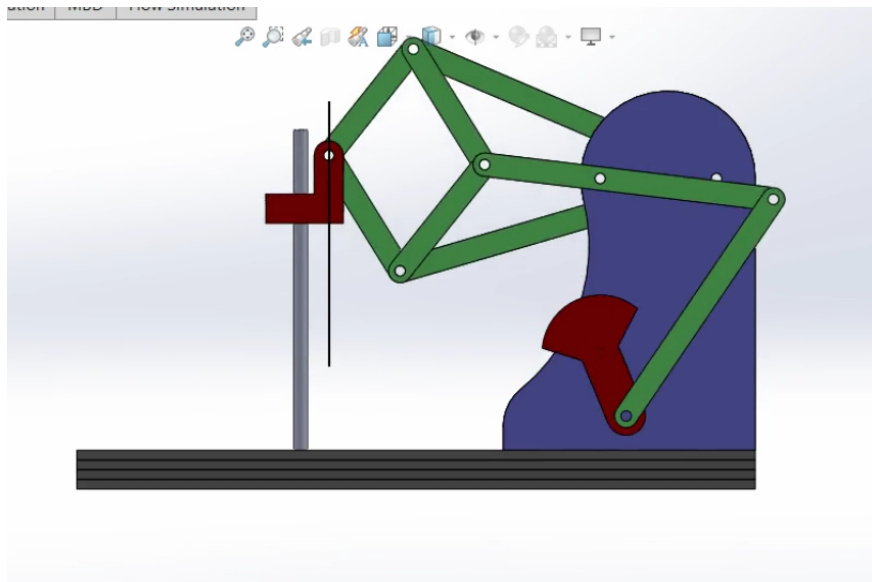


Figure 5: The Motion video for the linkage.

### 3 Kinematic and Kinetic Analysis

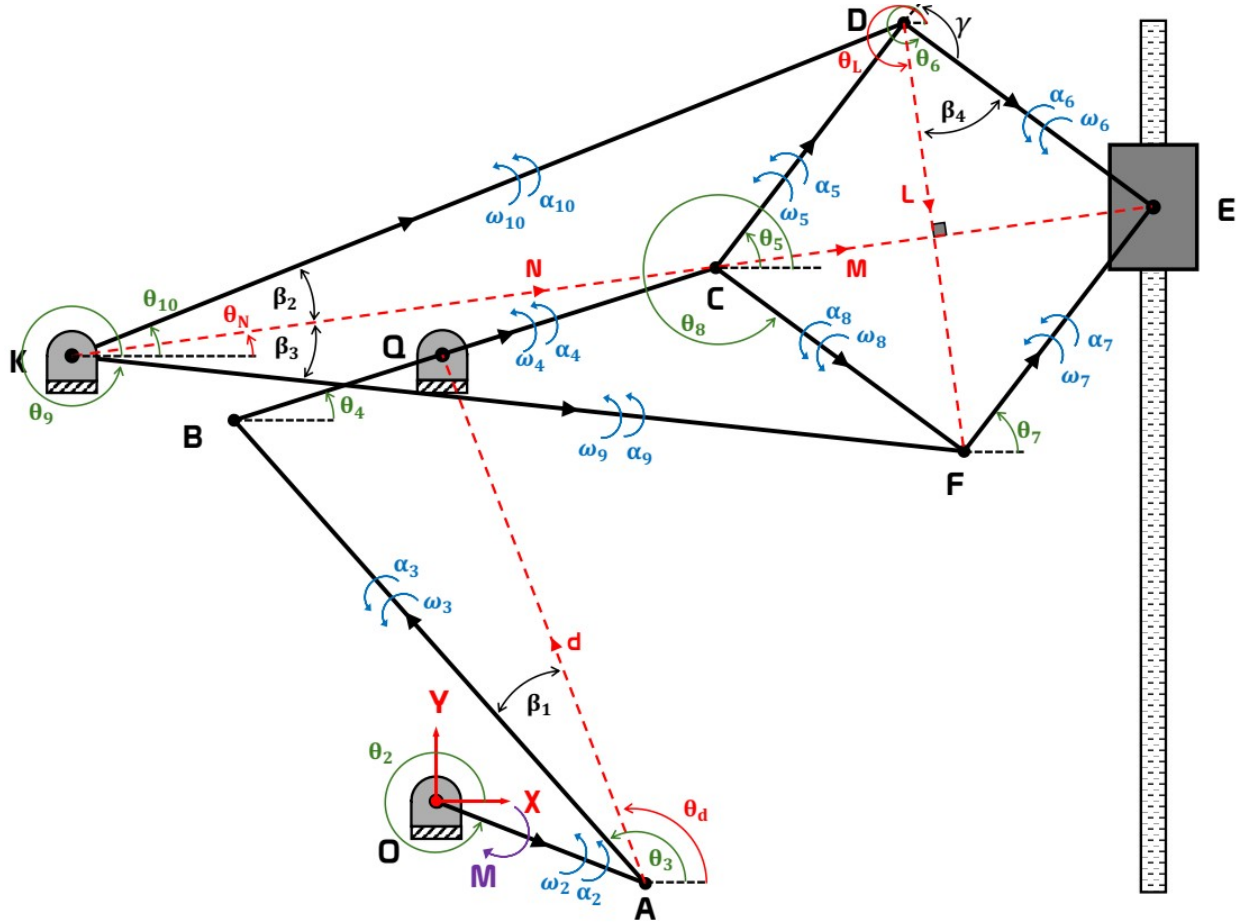


Figure 6: The Peaucellier-Lipkin Linkage.

The linkage shown in Fig. 6 consists of 10 links connected by 13 lower pairs and no higher pairs. The mobility of the mechanism is determined using the Gruebler equation:

**Gruebler equation**

$$M = 3(L - 1) - 2P_L - P_H$$

where  $M$  is the degree of freedom,  $L$  is the number of links,  $P_L$  is the number of lower pairs, and  $P_H$  is the number of higher pairs.

Substituting the corresponding values yields a degree of freedom equal to 1. This result indicates that one independent input parameter is required to completely define the configuration and motion of all links in the mechanism.

### 3.1 Position Analysis

To determine the configuration of the mechanism, the position analysis is performed by decomposing the linkage into 14 independent vector loops. Each link is represented as a position vector in the complex plane using Euler's identity:

#### Euler's identity

$$\vec{r} = r e^{i\theta}$$

Separating the resulting equations into real and imaginary components allows the Cartesian coordinates and angular positions of all joints to be obtained. Elimination techniques, such as multiplication by the complex conjugate, are employed to remove intermediate unknowns and solve the system sequentially, starting from the input crank and progressing toward the output point  $E$ .

#### 3.1.1 Loop Number One

$$\vec{OA} = x_A + iy_A = r_2 e^{i\theta_2} \quad (3.1.1)$$

(Real = Real)

$$\text{therefore } x_A = r_2 \cos(\theta_2) \quad (3.1.2)$$

(Imaginary = Imaginary)

$$\therefore y_A = r_2 \sin(\theta_2) \quad (3.1.3)$$

---

#### 3.1.2 Loop Number Two

$$\vec{OQ} = \vec{OA} + \vec{AQ} \quad (3.1.4)$$

$$x_Q + iy_Q = x_A + iy_A + d e^{i\theta_d} \quad (3.1.5)$$

$$d e^{i\theta_d} = (x_Q - x_A) + i(y_Q - y_A) \quad (3.1.6)$$

(Multiplying both sides by the conjugate)

$$d^2 = (x_Q - x_A)^2 + (y_Q - y_A)^2 \quad (3.1.7)$$

$$\therefore d = \sqrt{(x_Q - x_A)^2 + (y_Q - y_A)^2} \quad (3.1.8)$$

(Back to Eq. (3.1.6))

(Real = Real)

$$d \cos(\theta_d) = (x_Q - x_A) \quad (3.1.9)$$

$$\therefore \cos(\theta_d) = \frac{x_Q - x_A}{d} \quad (3.1.10)$$

(Imaginary = Imaginary)

$$d \sin(\theta_d) = (y_Q - y_A) \quad (3.1.11)$$

$$\therefore \sin(\theta_d) = \frac{y_Q - y_A}{d} \quad \therefore \theta_d \text{ is determined} \quad (3.1.12)$$


---

### 3.1.3 Loop Number Three

$$\vec{AQ} = \vec{AB} + \vec{BQ} \quad (3.1.13)$$

$$de^{i\theta_d} = r_3e^{i\theta_3} + X_4e^{i\theta_4} \quad (3.1.14)$$

$$X_4e^{i\theta_4} = (de^{i\theta_d} - r_3e^{i\theta_3}) \quad (3.1.15)$$

(Multiplying both sides by the conjugate)

$$X_4^2 = d^2 + r_3^2 - r_3de^{i(\theta_d-\theta_3)} - r_3de^{i(\theta_3-\theta_d)} \quad (3.1.16)$$

(assume that  $\beta = \theta_3 - \theta_d$ )

$$\therefore x_4^2 = d^2 + r_3^2 - r_3d(e^{-i\beta} + e^{i\beta}) \quad (3.1.17)$$

$$= d^2 + r_3^2 - 2r_3d \cos(\beta) \quad (3.1.18)$$

$$\therefore \cos(\beta) = \frac{X_4^2 - d^2 - r_3^2}{-2r_3d} \quad (3.1.19)$$

$$\therefore \beta = \cos^{-1} \left( \frac{X_4^2 - d^2 - r_3^2}{-2r_3d} \right) \quad (3.1.20)$$

$$\therefore \theta_3 = \beta + \theta_d \quad (3.1.21)$$

(Back to Eq. (3.1.15))

(Real = Real)

$$X_4 \cos(\theta_4) = d \cos(\theta_d) - r_3 \cos(\theta_3) \quad (3.1.22)$$

$$\cos(\theta_4) = \frac{d \cos(\theta_d) - r_3 \cos(\theta_3)}{X_4} \quad (3.1.23)$$

(Imaginary = Imaginary)

$$X_4 \sin(\theta_4) = d \sin(\theta_d) - r_3 \sin(\theta_3) \quad (3.1.24)$$

$$\sin(\theta_4) = \frac{d \sin(\theta_d) - r_3 \sin(\theta_3)}{X_4} \quad \therefore \theta_4 \text{ is determined} \quad (3.1.25)$$


---

#### 3.1.4 Loop Number Four

$$\vec{OB} = \vec{OA} + \vec{AB} \quad (3.1.26)$$

$$x_B + iy_B = x_A + iy_A + r_3 e^{i\theta_3} \quad (3.1.27)$$

(Real = Real)

$$\therefore x_B = x_A + r_3 \cos(\theta_3) \quad (3.1.28)$$

(Imaginary = Imaginary)

$$\therefore y_B = y_A + r_3 \sin(\theta_3) \quad (3.1.29)$$

---

#### 3.1.5 Loop Number five

$$\vec{OC} = \vec{OB} + \vec{BC} \quad (3.1.30)$$

$$x_C + iy_C = x_B + iy_B + r_4 e^{i\theta_4} \quad (3.1.31)$$

(Real = Real)

$$\therefore x_C = x_B + r_4 \cos(\theta_4) \quad (3.1.32)$$

(Imaginary = Imaginary)

$$\therefore y_C = y_B + r_4 \sin(\theta_4) \quad (3.1.33)$$

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#### 3.1.6 Loop Number six

$$\vec{OC} = \vec{OK} + \vec{KC} \quad (3.1.34)$$

$$x_C + iy_C = x_K + iy_K + N e^{i\theta_N} \quad (3.1.35)$$

$$N e^{i\theta_N} = (x_C - x_K) + i(y_C - y_K) \quad (3.1.36)$$

(Multiplying both sides by the conjugate)

$$N^2 = (x_C - x_K)^2 + (y_C - y_K)^2 \quad (3.1.37)$$

$$\therefore N = \sqrt{(x_C - x_K)^2 + (y_C - y_K)^2} \quad (3.1.38)$$

(Back to Eq. (3.1.36))



(Real = Real)

$$N \cos(\theta_N) = (x_C - x_K) + i(y_C - y_K) \quad (3.1.39)$$

$$\therefore \cos(\theta_N) = \frac{x_C - x_K}{N} \quad (3.1.40)$$

(Imaginary = Imaginary)

$$N \sin(\theta_N) = (y_C - y_K) + i(x_C - x_K) \quad (3.1.41)$$

$$\therefore \sin(\theta_N) = \frac{y_C - y_K}{N} \quad \therefore \theta_N \text{ is determined} \quad (3.1.42)$$

### 3.1.7 Loop Number Seven

$$\vec{KD} = \vec{KC} + \vec{CD} \quad (3.1.43)$$

$$r_{10}e^{i\theta_{10}} = Ne^{i\theta_N} + r_5e^{i\theta_5} \quad (3.1.44)$$

$$r_5e^{i\theta_5} = r_{10}e^{i\theta_{10}} - Ne^{i\theta_N} \quad (3.1.45)$$

(Multiplying both sides by the conjugate)

$$r_5^2 = r_{10}^2 + N^2 - 2r_{10}N \cos(\beta_2) \quad (3.1.46)$$

$$\therefore \cos(\beta_2) = \frac{N^2 + r_{10}^2 - r_5^2}{2r_{10}N} \quad \therefore \beta_2 \text{ is determined} \quad (3.1.47)$$

$$\therefore \theta_{10} = \beta_2 + \theta_N \quad (3.1.48)$$

(Real = Real)

$$\therefore \cos(\theta_5) = \frac{r_{10} \cos(\theta_{10}) - N \cos(\theta_N)}{r_5} \quad (3.1.49)$$

(Imaginary = Imaginary)

$$\therefore \sin(\theta_5) = \frac{r_{10} \sin(\theta_{10}) - N \sin(\theta_N)}{r_5} \quad \therefore \theta_5 \text{ is determined} \quad (3.1.50)$$

### 3.1.8 Loop Number Eight

$$\vec{KF} = \vec{KC} + \vec{CF} \quad (3.1.51)$$

$$r_9e^{i\theta_9} = Ne^{i\theta_N} + r_8e^{i\theta_8} \quad (3.1.52)$$

$$r_8e^{i\theta_8} = r_9e^{i\theta_9} - Ne^{i\theta_N} \quad (3.1.53)$$

(Multiplying both sides by the conjugate)

$$r_8^2 = r_9^2 + N^2 - 2r_9N \cos(\beta_3) \quad (3.1.54)$$

$$\therefore \cos(\beta_3) = \frac{r_9^2 + N^2 - r_8^2}{2r_9N} \quad (3.1.55)$$

$$\therefore \beta_3 = \cos^{-1}\left(\frac{r_9^2 + N^2 - r_8^2}{2r_9N}\right) \quad (3.1.56)$$

$$\therefore \theta_9 = \theta_N - \beta_3 \quad (3.1.57)$$

(Real = Real)

$$r_8 \cos(\theta_8) = r_9 \cos(i\theta_9) - N \cos(\theta_N) \quad (3.1.58)$$

$$\therefore \cos(\theta_8) = \frac{r_9 \cos(\theta_9) - N \cos(\theta_N)}{r_8} \quad (3.1.59)$$

(Imaginary = Imaginary)

$$r_8 \sin(\theta_8) = r_9 \sin(i\theta_9) - N \sin(\theta_N) \quad (3.1.60)$$

$$\therefore \sin(\theta_8) = \frac{r_9 \sin(\theta_9) - N \sin(\theta_N)}{r_8} \quad \therefore \theta_8 \text{ is determined} \quad (3.1.61)$$

### 3.1.9 Loop Number Nine

$$\vec{OF} = \vec{OK} + \vec{KF} \quad (3.1.62)$$

$$x_F + iy_F = x_K + iy_K + r_9 e^{i\theta_9} \quad (3.1.63)$$

(Real = Real)

$$\therefore x_F = x_K + r_9 \cos(\theta_9) \quad (3.1.64)$$

(Imaginary = Imaginary)

$$\therefore y_F = y_K + r_9 \sin(\theta_9) \quad (3.1.65)$$

### 3.1.10 Loop Number Ten

$$\vec{OD} = \vec{OK} + \vec{KD} \quad (3.1.66)$$

$$x_D + iy_D = x_K + iy_K + r_{10} e^{i\theta_{10}} \quad (3.1.67)$$

(Real = Real)

$$\therefore x_D = x_K + r_{10} \cos(\theta_{10}) \quad (3.1.68)$$

(Imaginary = Imaginary)

$$\therefore y_D = y_K + r_{10} \sin(\theta_{10}) \quad (3.1.69)$$

### 3.1.11 Loop Number Eleven

$$\vec{OF} = \vec{OD} + \vec{DF} \quad (3.1.70)$$

$$x_F + iy_F = x_D + iy_D + Le^{i\theta_L} \quad (3.1.71)$$

$$Le^{i\theta_L} = (x_F - x_D) + i(y_F - y_D) \quad (3.1.72)$$

(Multiplying both sides by the conjugate)

$$L^2 = (x_F - x_D)^2 + (y_F - y_D)^2 \quad (3.1.73)$$

$$\therefore L = \sqrt{(x_F - x_D)^2 + (y_F - y_D)^2} \quad (3.1.74)$$

(Back to Eq. (3.1.72))

(Real = Real)

$$L \cos(\theta_L) = x_F - x_D \quad (3.1.75)$$

$$\therefore \cos(\theta_L) = \frac{x_F - x_D}{L} \quad (3.1.76)$$

(Imaginary = Imaginary)

$$L \sin(\theta_L) = y_F - y_D \quad (3.1.77)$$

$$\therefore \sin(\theta_L) = \frac{y_F - y_D}{L} \quad \therefore \theta_L \text{ is determined} \quad (3.1.78)$$

### 3.1.12 Loop Number Twelve

$$\vec{DE} = \vec{DF} + \vec{FE} \quad (3.1.79)$$

$$r_6 e^{i\theta_6} = L e^{i\theta_L} + r_7 e^{i\theta_7} \quad (3.1.80)$$

$$r_7 e^{i\theta_7} = r_6 e^{i\theta_6} - L e^{i\theta_L} \quad (3.1.81)$$

(Multiplying both sides by the conjugate)

(assume that  $\beta_4 = \theta_6 - \theta_L$ )

$$r_7^2 = r_6^2 + L^2 - 2r_6 L \cos(\beta_4) \quad (3.1.82)$$

$$\therefore \cos(\beta_4) = \frac{r_6^2 + L^2 - r_7^2}{2r_6 L} \quad \therefore \beta_4 = \cos^{-1} \left( \frac{r_6^2 + L^2 - r_7^2}{2r_6 L} \right) \quad (3.1.83)$$

$$\therefore \theta_6 = \beta_4 + \theta_L \quad (3.1.84)$$

(Back to Eq. (3.1.81))

(Real = Real)

$$r_7 \cos(\theta_7) = r_6 \cos(\theta_6) - L \cos(\theta_L) \quad (3.1.85)$$

$$\therefore \cos(\theta_7) = \frac{r_6 \cos(\theta_6) - L \cos(\theta_L)}{r_7} \quad (3.1.86)$$

(Imaginary = Imaginary)

$$r_7 \sin(\theta_7) = r_6 \sin(\theta_6) - L \sin(\theta_L) \quad (3.1.87)$$

$$\therefore \sin(\theta_7) = \frac{r_6 \sin(\theta_6) - L \sin(\theta_L)}{r_7} \quad \therefore \theta_7 \text{ is determined} \quad (3.1.88)$$


---

### 3.1.13 Loop Number Thirteen

$$\vec{CE} = \vec{CD} + \vec{DE} \quad (3.1.89)$$

$$Me^{i\theta_M} = r_5 e^{i\theta_5} + r_6 e^{i\theta_6} \quad (3.1.90)$$

(Multiplying both sides by the conjugate)

$$M^2 = r_5^2 + r_6^2 + 2r_5 r_6 \cos(\gamma) \quad (3.1.91)$$

$$\therefore M = \sqrt{r_5^2 + r_6^2 + 2r_5 r_6 \cos(\gamma)} \quad (3.1.92)$$

(Alternative form)

$$\theta_5 - \theta_6 = \gamma \quad (3.1.93)$$

$$\gamma + 2\beta_4 = 180 \quad (3.1.94)$$

$$2\beta_4 = 180 - \gamma \quad (3.1.95)$$

$$-\cos(2\beta_4) = \cos(\gamma) \quad (3.1.96)$$

(Back to Eq. (3.1.91) and modifying the angle)

$$M^2 = r_5^2 + r_6^2 - 2r_5 r_6 \cos(2\beta_4) \quad (3.1.97)$$

$$M = \sqrt{r_5^2 + r_6^2 - 2r_5 r_6 \cos(2\beta_4)} \quad (3.1.98)$$


---

### 3.1.14 Loop Number Fourteen

$$\vec{OE} = \vec{OC} + \vec{CE} \quad (3.1.99)$$

$$x_E + iy_E = x_C + iy_C + Me^{i\theta_M} \quad (3.1.100)$$

(Real = Real)

$$\therefore x_E = x_C + M \cos(\theta_M) \quad (3.1.101)$$

(Imaginary = Imaginary)

$$\therefore y_E = y_C + M \sin(\theta_M) \quad (3.1.102)$$


---

## 3.2 Velocity Analysis

The velocity analysis is conducted to determine the angular velocities of all links and the linear velocities of the joints, as illustrated in Fig. 6. Fixed points  $O$ ,  $Q$ , and  $K$  are assumed to have zero velocity, while the input angular velocity  $\omega_2$  is prescribed.

### Starting Point

$$\vec{V}_A = i\omega_2 r_2 e^{i\theta_2}$$

### Relative Velocity

$$\vec{V}_B = \vec{V}_A + \vec{V}_{BA}$$

The analysis begins with the velocity of point  $A$ , which is directly driven by the input crank. Velocities of subsequent joints are obtained using relative velocity relationships. By separating the resulting equations into real and imaginary components, the unknown angular and linear velocities are solved sequentially. The analysis concludes at the output point  $E$ , where the vertical guide constraint imposes a zero horizontal velocity component.

### 3.2.1 Loop Number One

$$\vec{V}_A = \vec{V}_O + \vec{V}_{AO} \quad (\vec{V}_O = 0.0) \quad (3.2.1)$$

$$V_A^x + iV_A^y = 0 + i\omega_2 r_2 e^{i\theta_2} \quad (3.2.2)$$

(Real = Real)

$$\therefore V_A^x = -\omega_2 r_2 \sin(\theta_2) \quad (3.2.3)$$

(Imaginary = Imaginary)

$$\therefore V_A^y = \omega_2 r_2 \cos(\theta_2) \quad (3.2.4)$$

### 3.2.2 Loop Number Two

$$\vec{V}_B = \vec{V}_Q + \vec{V}_{BQ} \quad \text{and} \quad \vec{V}_B = \vec{V}_A + \vec{V}_{BA} \quad (3.2.5)$$

$$\vec{V}_Q + \vec{V}_{BQ} = \vec{V}_A + \vec{V}_{BA} \quad (\vec{V}_Q = 0) \quad (3.2.6)$$

$$-i\omega_4 x_4 e^{i\theta_4} = V_A^x + iV_A^y + i\omega_3 r_3 e^{i\theta_3} \quad (3.2.7)$$

(Multiplying both sides by  $e^{-i\theta_4}$ )

$$-i\omega_4 x_4 = (V_A^x + iV_A^y)e^{-i\theta_4} + i\omega_3 r_3 e^{i(\theta_3 - \theta_4)} \quad (3.2.8)$$

(Real = Real)

$$\therefore 0 = V_A^x \cos(\theta_4) + V_A^y \sin(\theta_4) - \omega_3 r_3 \sin(\theta_3 - \theta_4) \quad (3.2.9)$$

$$\therefore \omega_3 = \frac{V_A^x \cos(\theta_4) + V_A^y \sin(\theta_4)}{r_3 \sin(\theta_3 - \theta_4)} \quad (3.2.10)$$

(Imaginary = Imaginary)

$$\therefore -\omega_4 x_4 = -V_A^x \sin(\theta_4) + V_A^y \cos(\theta_4) + \omega_3 r_3 \cos(\theta_3 - \theta_4) \quad (3.2.11)$$

$$\omega_4 = \frac{V_A^x \sin(\theta_4) - V_A^y \cos(\theta_4) - \omega_3 r_3 \cos(\theta_3 - \theta_4)}{X_4} \quad (3.2.12)$$

(Back to Eq. (3.2.5))

$$\vec{V}_B = \vec{V}_A + \vec{V}_{BA} \quad (3.2.13)$$

$$V_B^x + iV_B^y = V_A^x + iV_A^y + i\omega_3 r_3 e^{i\theta_3} \quad (3.2.14)$$

(Real = Real)

$$\therefore V_B^x = V_A^x - \omega_3 r_3 \sin(\theta_3) \quad (3.2.15)$$

(Imaginary = Imaginary)

$$\therefore V_B^y = V_A^y + \omega_3 r_3 \cos(\theta_3) \quad (3.2.16)$$

### 3.2.3 Loop Number Three

$$\vec{V}_C = \vec{V}_Q + \vec{V}_{CQ} \quad (\vec{V}_Q = 0.0) \quad (3.2.17)$$

$$V_C^x + iV_C^y = 0 + i\omega_4(r_4 - X_4)e^{i\theta_4} \quad (3.2.18)$$

(Real = Real)

$$\therefore V_C^x = -\omega_4(r_4 - X_4) \sin(\theta_4) \quad (3.2.19)$$

(Imaginary = Imaginary)

$$\therefore V_C^y = \omega_4(r_4 - X_4) \cos(\theta_4) \quad (3.2.20)$$

### 3.2.4 Loop Number Four

$$\vec{V}_f = \vec{V}_K + \vec{V}_{fK} \quad \text{and} \quad \vec{V}_f = \vec{V}_C + \vec{V}_{fC} \quad (3.2.21)$$

$$\vec{V}_f = \vec{V}_K + \vec{V}_{fK} \quad (\vec{V}_K = 0.0) \quad (3.2.22)$$

$$V_f^x + iV_f^y = 0 + i\omega_9 r_9 e^{i\theta_9} \quad (3.2.23)$$

(Real = Real)

$$\therefore V_f^x = -\omega_9 r_9 \sin(\theta_9) \quad (3.2.24)$$

(Imaginary = Imaginary)

$$\therefore V_f^y = \omega_9 r_9 \cos(\theta_9) \quad (3.2.25)$$

(Back to Eq. (3.2.21) part 2)

$$i\omega_9 r_9 e^{i\theta_9} = V_C^x + iV_C^y + i\omega_8 r_8 e^{i\theta_8} \quad (3.2.26)$$

(Multiplying both sides by  $e^{-i\theta_9}$ )

$$(3.2.27)$$

$$i\omega_9 r_9 = (V_C^x + iV_C^y)e^{-i\theta_9} + i\omega_8 r_8 e^{i(\theta_8 - \theta_9)} \quad (3.2.28)$$

(Real = Real)

$$0 = V_C^x \cos(\theta_9) + V_C^y \sin(\theta_9) - \omega_8 r_8 \sin(\theta_8 - \theta_9) \quad (3.2.29)$$

$$\therefore \omega_8 = \frac{V_C^x \cos(\theta_9) + V_C^y \sin(\theta_9)}{r_8 \sin(\theta_8 - \theta_9)} \quad (3.2.30)$$

(Imaginary = Imaginary)

$$\omega_9 r_9 = -V_C^x \sin(\theta_9) + V_C^y \cos(\theta_9) + \omega_8 r_8 \cos(\theta_8 - \theta_9) \quad (3.2.31)$$

$$\therefore \omega_9 = \frac{-V_C^x \sin(\theta_9) + V_C^y \cos(\theta_9) + \omega_8 r_8 \cos(\theta_8 - \theta_9)}{r_9} \quad (3.2.32)$$

### 3.2.5 Loop Number Five

$$\vec{V}_D = \vec{V}_C + \vec{V}_{DC} \quad (3.2.33)$$

$$i\omega_{10}r_{10}e^{i\theta_{10}} = V_C^x + iV_C^y + i\omega_5r_5e^{i\theta_5} \quad (3.2.34)$$

(Multiplying both sides by  $e^{-i\theta_{10}}$ )

$$i\omega_{10}r_{10} = (V_C^x + iV_C^y)e^{-i\theta_{10}} + i\omega_5r_5e^{i(\theta_5-\theta_{10})} \quad (3.2.35)$$

(Real = Real)

$$0 = V_C^x \cos(\theta_{10}) + V_C^y \sin(\theta_{10}) - \omega_5r_5 \sin(\theta_5 - \theta_{10}) \quad (3.2.36)$$

$$\therefore \omega_5 = \frac{V_C^x \cos(\theta_{10}) + V_C^y \sin(\theta_{10})}{r_5 \sin(\theta_5 - \theta_{10})} \quad (3.2.37)$$

(Imaginary = Imaginary)

$$\omega_{10}r_{10} = -V_C^x \sin(\theta_{10}) + V_C^y \cos(\theta_{10}) + \omega_5r_5 \cos(\theta_5 - \theta_{10}) \quad (3.2.38)$$

$$\therefore \omega_{10} = \frac{-V_C^x \sin(\theta_{10}) + V_C^y \cos(\theta_{10}) + \omega_5r_5 \cos(\theta_5 - \theta_{10})}{r_{10}} \quad (3.2.39)$$


---

### 3.2.6 Loop Number Six

$$\vec{V}_D = \vec{V}_K + \vec{V}_{DK} \quad (3.2.40)$$

$$V_D^x + iV_D^y = i\omega_{10}r_{10}e^{i\theta_{10}} \quad (3.2.41)$$

(Real = Real)

$$\therefore V_D^x = -\omega_{10}r_{10} \sin(\theta_{10}) \quad (3.2.42)$$

(Imaginary = Imaginary)

$$\therefore V_D^y = \omega_{10}r_{10} \cos(\theta_{10}) \quad (3.2.43)$$


---



### 3.2.7 Loop Number Seven

$$\vec{V}_E = \vec{V}_D + \vec{V}_{ED} \quad (3.2.44)$$

$$V_E^x + iV_E^y = V_D^x + iV_D^y + i\omega_6 r_6 e^{i\theta_6} \quad (3.2.45)$$

(Real = Real)

$$V_E^x = V_D^x - \omega_6 r_6 \sin(\theta_6) \quad (V_E^x = 0.0) \quad (3.2.46)$$

$$0 = V_D^x - \omega_6 r_6 \sin(\theta_6) \quad (3.2.47)$$

$$\therefore \omega_6 = \frac{V_D^x}{r_6 \sin(\theta_6)} \quad (3.2.48)$$

(Imaginary = Imaginary)

$$\therefore V_E^y = V_D^y + \omega_6 r_6 \cos(\theta_6) \quad (3.2.49)$$

### 3.2.8 Loop Number Eight

$$\vec{V}_E = \vec{V}_F + \vec{V}_{EF} \quad (3.2.50)$$

$$V_E^x + iV_E^y = V_F^x + iV_F^y + i\omega_7 r_7 e^{i\theta_7} \quad (3.2.51)$$

(Real = Real)

$$V_E^x = V_F^x - \omega_7 r_7 \sin(\theta_7) \quad (V_E^x = 0.0) \quad (3.2.52)$$

$$0 = V_F^x - \omega_7 r_7 \sin(\theta_7) \quad (3.2.53)$$

$$\therefore \omega_7 = \frac{V_F^x}{r_7 \sin(\theta_7)} \quad (3.2.54)$$

(Imaginary = Imaginary)

$$\therefore V_E^y = V_F^y + \omega_7 r_7 \cos(\theta_7) \quad (3.2.55)$$

### 3.3 Acceleration Analysis

The acceleration analysis represents the final stage of the kinematic study and aims to determine the angular accelerations of the links and the linear accelerations of the joints, as shown in Fig. 6. Similar to the previous analyses, vector loop equations are formulated for each loop of the mechanism.

For a general link, the acceleration vector is expressed as:

#### Starting Point

$$\vec{A} = (-\omega^2 + i\alpha)re^{i\theta}$$

#### Relative Acceleration

$$\vec{A}_B = \vec{A}_A + \vec{A}_{BA}$$

The analysis begins with the input crank  $OA$ . Since the input angular velocity  $\omega_2$  is assumed constant, the corresponding angular acceleration  $\alpha_2$  is taken as zero. By separating the equations into real and imaginary components, the unknown accelerations are determined sequentially. The analysis terminates at the slider point  $E$ , where the horizontal acceleration component is constrained to zero.

#### 3.3.1 Loop Number One

$$\vec{A}_A = \vec{A}_O + \vec{A}_{AO} \quad (\vec{A}_O = 0) \quad (3.3.1)$$

$$A_A^x + iA_A^y = (-\omega_2^2 + i\alpha_2)r_2e^{i\theta_2} \quad (\alpha_2 = 0) \quad (3.3.2)$$

(Real = Real)

$$\therefore A_A^x = -\omega_2^2 r_2 \cos(\theta_2) \quad (3.3.3)$$

(Imaginary = Imaginary)

$$\therefore A_A^y = -\omega_2^2 r_2 \sin(\theta_2) \quad (3.3.4)$$

#### 3.3.2 Loop Number Two

$$\vec{A}_B = \vec{A}_A + \vec{A}_{BA} \quad (3.3.5)$$

$$(-\omega_4^2 + i\alpha_4)(-X_4)e^{i\theta_4} = A_A^x + iA_A^y + (-\omega_3^2 + i\alpha_3)r_3e^{i\theta_3} \quad (3.3.6)$$

(Multiplying both sides by  $e^{-i\theta_4}$ )

$$(\omega_4^2 - i\alpha_4)X_4 = (A_A^x + iA_A^y)e^{-i\theta_4} + (-\omega_3^2 + i\alpha_3)r_3e^{i(\theta_3-\theta_4)} \quad (3.3.7)$$

(Real = Real)

$$\omega_4^2 X_4 = A_A^x \cos \theta_4 + A_A^y \sin \theta_4 - \omega_3^2 r_3 \cos(\theta_3 - \theta_4) - \alpha_3 r_3 \sin(\theta_3 - \theta_4) \quad (3.3.8)$$

$$\therefore \alpha_3 = \frac{A_A^x \cos \theta_4 + A_A^y \sin \theta_4 - \omega_3^2 r_3 \cos(\theta_3 - \theta_4) - \omega_4^2 X_4}{r_3 \sin(\theta_3 - \theta_4)} \quad (3.3.9)$$

(Imaginary = Imaginary)

$$-\alpha_4 X_4 = -A_A^x \sin \theta_4 + A_A^y \cos \theta_4 - \omega_3^2 r_3 \sin(\theta_3 - \theta_4) + \alpha_3 r_3 \cos(\theta_3 - \theta_4) \quad (3.3.10)$$

$$\therefore \alpha_4 = \frac{A_A^x \sin \theta_4 - A_A^y \cos \theta_4 + \omega_3^2 r_3 \sin(\theta_3 - \theta_4) - \alpha_3 r_3 \cos(\theta_3 - \theta_4)}{X_4} \quad (3.3.11)$$

### 3.3.3 Loop Number Three

$$\vec{A}_C = \vec{A}_Q + \vec{A}_{CQ} \quad (3.3.12)$$

$$A_C^x + iA_C^y = (-\omega_4^2 + i\alpha_4)(r_4 - X_4)e^{i\theta_4} \quad (3.3.13)$$

(Real = Real)

$$\therefore A_C^x = -\omega_4^2(r_4 - X_4) \cos(\theta_4) - \alpha_4(r_4 - X_4) \sin(\theta_4) \quad (3.3.14)$$

(Imaginary = Imaginary)

$$\therefore A_C^y = -\omega_4^2(r_4 - X_4) \sin(\theta_4) + \alpha_4(r_4 - X_4) \cos(\theta_4) \quad (3.3.15)$$

### 3.3.4 Loop Number Four

$$\vec{A}_F = \vec{A}_C + \vec{A}_{FC} \quad (3.3.16)$$

$$(-\omega_9^2 + i\alpha_9)r_9e^{i\theta_9} = A_C^x + iA_C^y + (-\omega_8^2 + i\alpha_8)r_8e^{i\theta_8} \quad (3.3.17)$$

(Multiplying both sides by  $e^{-i\theta_9}$ )

$$(-\omega_9^2 + i\alpha_9)r_9 = (A_C^x + iA_C^y)e^{-i\theta_9} + (-\omega_8^2 + i\alpha_8)r_8e^{i(\theta_8-\theta_9)} \quad (3.3.18)$$

(Real = Real)

$$-\omega_9^2 r_9 = A_C^x \cos(\theta_9) + A_C^y \sin(\theta_9) - \omega_8^2 r_8 \cos(\theta_8 - \theta_9) - \alpha_8 r_8 \sin(\theta_8 - \theta_9) \quad (3.3.19)$$

$$\therefore \alpha_8 = \frac{A_C^x \cos(\theta_9) + A_C^y \sin(\theta_9) - \omega_8^2 r_8 \cos(\theta_8 - \theta_9) + \omega_9^2 r_9}{r_8 \sin(\theta_8 - \theta_9)} \quad (3.3.20)$$

(Imaginary = Imaginary)

$$\alpha_9 r_9 = -A_C^x \sin(\theta_9) + A_C^y \cos(\theta_9) - \omega_8^2 r_8 \sin(\theta_8 - \theta_9) + \alpha_8 r_8 \cos(\theta_8 - \theta_9) \quad (3.3.21)$$

$$\therefore \alpha_9 = \frac{-A_C^x \sin(\theta_9) + A_C^y \cos(\theta_9) - \omega_8^2 r_8 \sin(\theta_8 - \theta_9) - \alpha_8 r_8 \cos(\theta_8 - \theta_9)}{r_9} \quad (3.3.22)$$


---

### 3.3.5 Loop Number Five

$$\vec{A}_D = \vec{A}_C + \vec{A}_{DC} \quad (3.3.23)$$

$$(-\omega_{10}^2 + i\alpha_{10})r_{10}e^{i\theta_{10}} = A_C^x + iA_C^y + (-\omega_5^2 + i\alpha_5)r_5e^{i\theta_5} \quad (3.3.24)$$

(Multiplying both sides by  $e^{-i\theta_{10}}$ )

$$(-\omega_{10}^2 + i\alpha_{10})r_{10} = (A_C^x + iA_C^y)e^{-i\theta_{10}} + (-\omega_5^2 + i\alpha_5)r_5e^{i(\theta_5 - \theta_{10})} \quad (3.3.25)$$

(Real = Real)

$$-\omega_{10}^2 r_{10} = A_C^x \cos(\theta_{10}) + A_C^y \sin(\theta_{10}) - \omega_5^2 r_5 \cos(\theta_5 - \theta_{10}) - \alpha_5 r_5 \sin(\theta_5 - \theta_{10}) \quad (3.3.26)$$

$$\therefore \alpha_5 = \frac{A_C^x \cos(\theta_{10}) + A_C^y \sin(\theta_{10}) - \omega_5^2 r_5 \cos(\theta_5 - \theta_{10}) + \omega_{10}^2 r_{10}}{r_5 \sin(\theta_5 - \theta_{10})} \quad (3.3.27)$$

(Imaginary = Imaginary)

$$\alpha_{10} r_{10} = -A_C^x \sin(\theta_{10}) + A_C^y \cos(\theta_{10}) - \omega_5^2 r_5 \sin(\theta_5 - \theta_{10}) + \alpha_5 r_5 \cos(\theta_5 - \theta_{10}) \quad (3.3.28)$$

$$\therefore \alpha_{10} = \frac{-A_C^x \sin(\theta_{10}) + A_C^y \cos(\theta_{10}) - \omega_5^2 r_5 \sin(\theta_5 - \theta_{10}) + \alpha_5 r_5 \cos(\theta_5 - \theta_{10})}{r_{10}} \quad (3.3.29)$$


---

### 3.3.6 Loop Number Six

$$\vec{A}_D = \vec{A}_K + \vec{A}_{DK} \quad (3.3.30)$$

$$A_D^x + iA_D^y = (-\omega_{10}^2 + i\alpha_{10})r_{10}e^{i\theta_{10}} \quad (3.3.31)$$

(Real = Real)

$$\therefore A_D^x = -\omega_{10}^2 r_{10} \cos(\theta_{10}) - \alpha_{10} r_{10} \sin(\theta_{10}) \quad (3.3.32)$$

(Imaginary = Imaginary)

$$\therefore A_D^y = -\omega_{10}^2 r_{10} \sin(\theta_{10}) + \alpha_{10} r_{10} \cos(\theta_{10}) \quad (3.3.33)$$


---

### 3.3.7 Loop Number Seven

$$\vec{A}_F = \vec{A}_K + \vec{A}_{FK} \quad (3.3.34)$$

$$A_F^x + iA_F^y = (-\omega_9^2 + i\alpha_9)r_9e^{i\theta_9} \quad (3.3.35)$$

(Real = Real)

$$\therefore A_F^x = -\omega_9^2 r_9 \cos(\theta_9) - \alpha_9 r_9 \sin(\theta_9) \quad (3.3.36)$$

(Imaginary = Imaginary)

$$\therefore A_F^y = -\omega_9^2 r_9 \sin(\theta_9) + \alpha_9 r_9 \cos(\theta_9) \quad (3.3.37)$$


---

### 3.3.8 Loop Number Eight

$$\vec{A}_E = \vec{A}_D + \vec{A}_{ED} \quad (3.3.38)$$

$$A_E^x + iA_E^y = A_D^x + iA_D^y + (-\omega_6^2 + i\alpha_6)r_6e^{i\theta_6} \quad (3.3.39)$$

(Real = Real)

$$0 = A_D^x - \omega_6^2 r_6 \cos(\theta_6) - \alpha_6 r_6 \sin(\theta_6) \quad (A_E^x = 0.0) \quad (3.3.40)$$

$$\therefore \alpha_6 = \frac{A_D^x - \omega_6^2 r_6 \cos(\theta_6)}{r_6 \sin(\theta_6)} \quad (3.3.41)$$

(Imaginary = Imaginary)

$$\therefore A_E^y = A_D^y - \omega_6^2 r_6 \sin(\theta_6) + \alpha_6 r_6 \cos(\theta_6) \quad (3.3.42)$$

$$(3.3.43)$$


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### 3.3.9 Loop Number Nine

$$\vec{A}_E = \vec{A}_F + \vec{A}_{EF} \quad (3.3.44)$$

$$A_E^x + iA_E^y = A_F^x + iA_F^y + (-\omega_7^2 + i\alpha_7)r_7e^{i\theta_7} \quad (3.3.45)$$

(Real = Real)

$$0 = A_F^x - \omega_7^2 r_7 \cos(\theta_7) - \alpha_7 r_7 \sin(\theta_7) \quad (A_E^x = 0.0) \quad (3.3.46)$$

$$\therefore \alpha_7 = \frac{A_F^x - \omega_7^2 r_7 \cos(\theta_7)}{r_7 \sin(\theta_7)} \quad (3.3.47)$$


---

### 3.4 Force Analysis

The force analysis is performed to determine the unknown joint forces and reaction forces acting on the links of the mechanism. This analysis is carried out using Newton's laws of motion in conjunction with D'Alembert's principle, in which the dynamic system is converted into an equivalent static system by introducing inertia forces and inertia moments. This transformation allows the equilibrium equations:

**Newton's laws of motion in conjunction with D'Alembert's principle**

$$\sum F = 0 \quad \text{and} \quad \sum M = 0$$

to be applied to each link to evaluate the internal and external forces.

#### 3.4.1 Link Number Five and Six

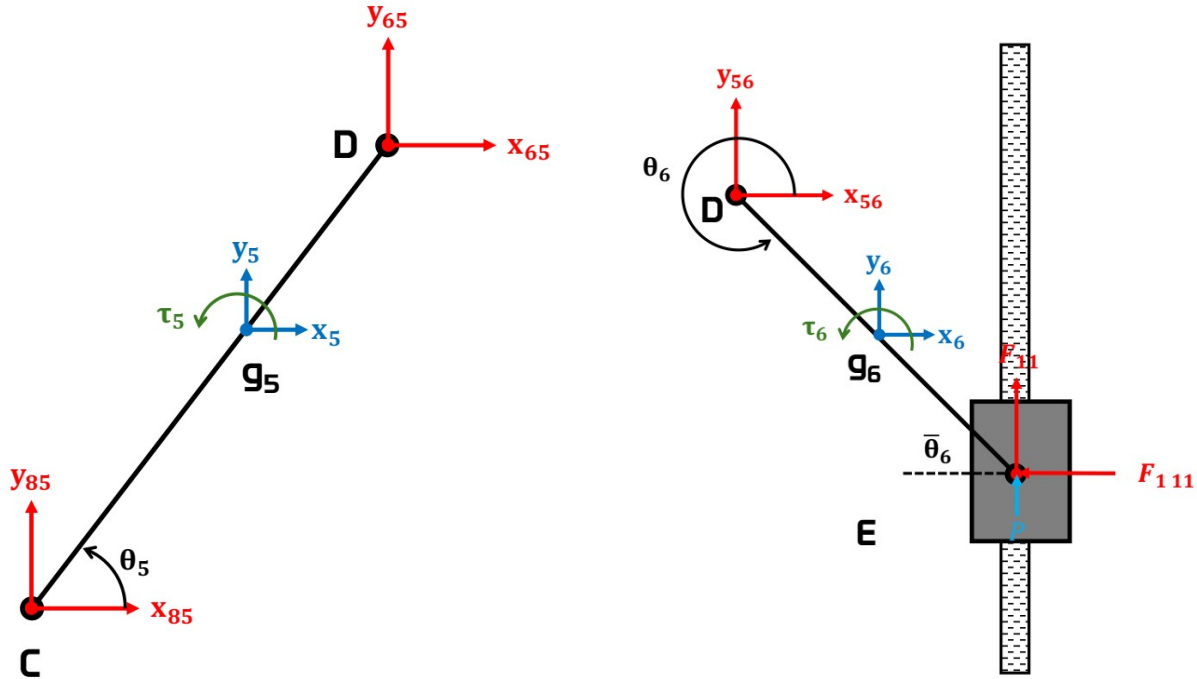


Figure 7: The free body diagram of Links 5, (6 + slider)

$$\begin{bmatrix} x_5 = -m_5 A_{G_5}^x \\ y_5 = -m_5 A_{G_5}^y \\ T_5 = -I_5 \alpha_5 \end{bmatrix} \quad \begin{bmatrix} \dot{\theta}_6 = 360 - \theta_6 \\ x_6 = -m_6 A_{G_6}^x \\ y_6 = -m_6 A_{G_6}^y \\ T_6 = -I_6 \alpha_6 \end{bmatrix} \quad \begin{bmatrix} \dot{\theta}_6 = 360 - \theta_6 \\ F_{11} = -m_{11} A_{G_{11}}^y \\ T_6 = -I_6 \alpha_6 \end{bmatrix}$$

$$\sum M_E = 0$$

$$\therefore -x_{(5-6)}r_6 \sin(\dot{\theta}_6) - x_6 \frac{r_6}{2} \sin(\dot{\theta}_6) - y_{(5-6)}r_6 \cos(\dot{\theta}_6) - y_6 \frac{r_6}{2} \cos(\dot{\theta}_6) + T_6 = 0 \quad (3.4.1)$$

$$\therefore x_{(5-6)} = \frac{x_6 \frac{r_6}{2} \sin(\dot{\theta}_6) + y_{(5-6)}r_6 \cos(\dot{\theta}_6) + y_6 \frac{r_6}{2} \cos(\dot{\theta}_6) - T_6}{-r_6 \sin(\dot{\theta}_6)}$$

$$\sum M_C = 0$$

$$\therefore -x_{(6-5)}r_5 \sin(\theta_5) - x_5 \frac{r_5}{2} \sin(\theta_5) + y_{(6-5)}r_6 \cos(\theta_5) + y_5 \frac{r_5}{2} \cos(\theta_5) + T_5 = 0 \quad (3.4.2)$$

$$\therefore x_{(6-5)} = \frac{x_5 \frac{r_5}{2} \sin(\theta_5) - y_{(6-5)}r_6 \cos(\theta_5) - y_5 \frac{r_5}{2} \cos(\theta_5) - T_5}{-r_5 \sin(\theta_5)}$$

$$\sum M_D = 0$$

$$\therefore (F_{11} + P)r_6 \cos(\dot{\theta}_6) + y_6 \frac{r_6}{2} \cos(\dot{\theta}_6) + x_6 \frac{r_6}{2} \sin(\dot{\theta}_6) + T_6 - F_{(1-11)}r_6 \sin(\dot{\theta}_6) = 0 \quad (3.4.3)$$

$$\therefore F_{(1-11)} = \frac{(F_{11} + P)r_6 \cos(\dot{\theta}_6) + y_6 \frac{r_6}{2} \cos(\dot{\theta}_6) + x_6 \frac{r_6}{2} \sin(\dot{\theta}_6) - T_6}{r_6 \sin(\dot{\theta}_6)}$$

(By adding both Eq. (3.4.1) and Eq. (3.4.2))

(By applying sigma forces on Link 5)

$$\begin{array}{ll} \sum F_y = 0 & \sum F_x = 0 \\ y_{(6-5)} + y_5 + y_{(8-5)} = 0 & x_{(6-5)} + x_5 + x_{(8-5)} = 0 \\ \therefore y_{(8-5)} = -y_{(6-5)} - y_5 & \therefore x_{(8-5)} = -x_{(6-5)} - x_5 \end{array}$$

(By applying sigma forces on Link 6)

$$\begin{array}{ll} \sum F_y = 0 & \sum F_x = 0 \\ y_{(5-6)} + y_6 + y_{(7-6)} = 0 & x_{(5-6)} + x_6 + x_{(7-6)} = 0 \\ \therefore y_{(7-6)} = -y_{(5-6)} - y_6 & \therefore x_{(7-6)} = x_{(5-6)} - x_6 \end{array}$$


---

### 3.4.2 Link Number Seven and Eight

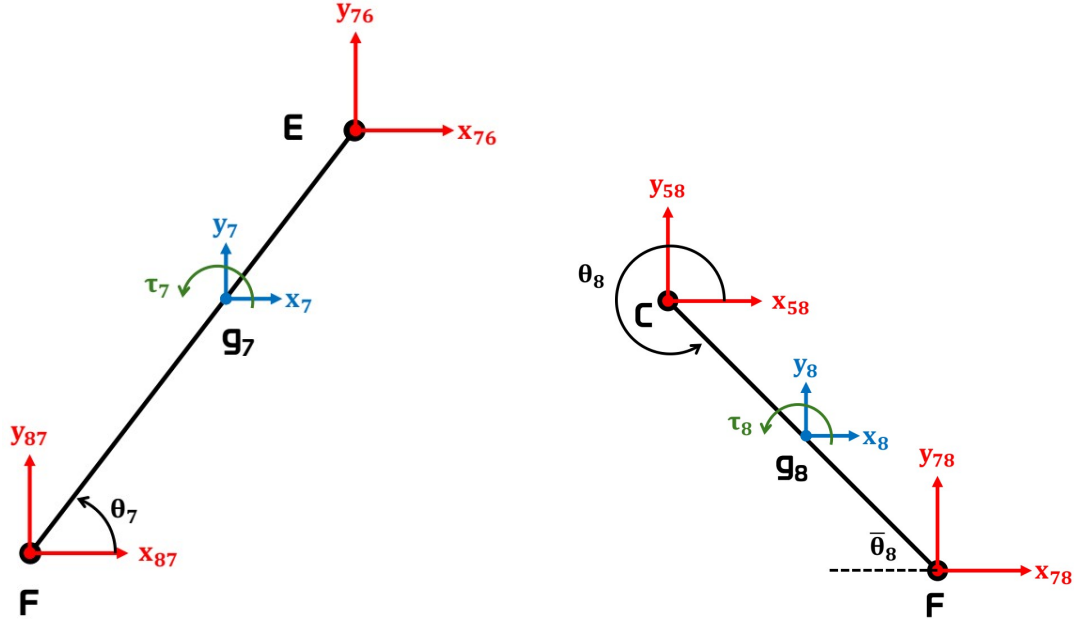


Figure 8: The free body diagram of Links 7, 8

$$\begin{bmatrix} x_7 = -m_7 A_{G_7}^x \\ y_7 = -m_7 A_{G_7}^y \\ T_7 = -I_7 \alpha_7 \end{bmatrix} \quad \begin{bmatrix} \dot{\theta}_8 = 360 - \theta_8 \\ x_8 = -m_8 A_{G_8}^x \\ y_8 = -m_8 A_{G_8}^y \\ T_8 = -I_8 \alpha_8 \end{bmatrix}$$

$$\sum M_C = 0$$

$$\therefore x_{(7-8)} r_8 \sin(\dot{\theta}_8) + x_8 \frac{r_8}{2} \sin(\dot{\theta}_8) + y_{(7-8)} r_8 \cos(\dot{\theta}_8) + y_8 \frac{r_8}{2} \cos(\dot{\theta}_8) + T_8 = 0 \quad (3.4.4)$$

$$\sum M_E = 0$$

$$\therefore x_{(8-7)} r_7 \sin(\theta_7) + x_7 \frac{r_7}{2} \sin(\theta_7) - y_{(8-7)} r_7 \cos(\theta_7) - y_7 \frac{r_7}{2} \cos(\theta_7) + T_7 = 0 \quad (3.4.5)$$

(By adding both Eq. (3.4.4) and Eq. (3.4.5))

$$\therefore x_{(7-8)} \quad \text{and} \quad y_{(7-8)} \quad \text{are determined}$$

(By applying sigma forces on Link 7)

$$\begin{array}{ll} \sum F_y = 0 & \sum F_x = 0 \\ y_{(6-7)} + y_7 + y_{(8-7)} = 0 & x_{(6-7)} + x_7 + x_{(8-7)} = 0 \\ \therefore y_{(6-7)} = -y_7 - y_{(8-7)} & \therefore x_{(6-7)} = -x_7 - x_{(8-7)} \end{array}$$



(By applying sigma forces on Link 8)

$$\begin{aligned}\sum F_y &= 0 \\ y_{(5-8)} + y_8 + y_{(7-8)} &= 0 \\ \therefore y_{(5-8)} &= -y_8 - y_{(7-8)}\end{aligned}$$

$$\begin{aligned}\sum F_x &= 0 \\ x_{(5-8)} + x_8 + x_{(7-8)} &= 0 \\ \therefore x_{(5-8)} &= -x_8 - x_{(7-8)}\end{aligned}$$


---

### 3.4.3 Links Number Nine and Ten

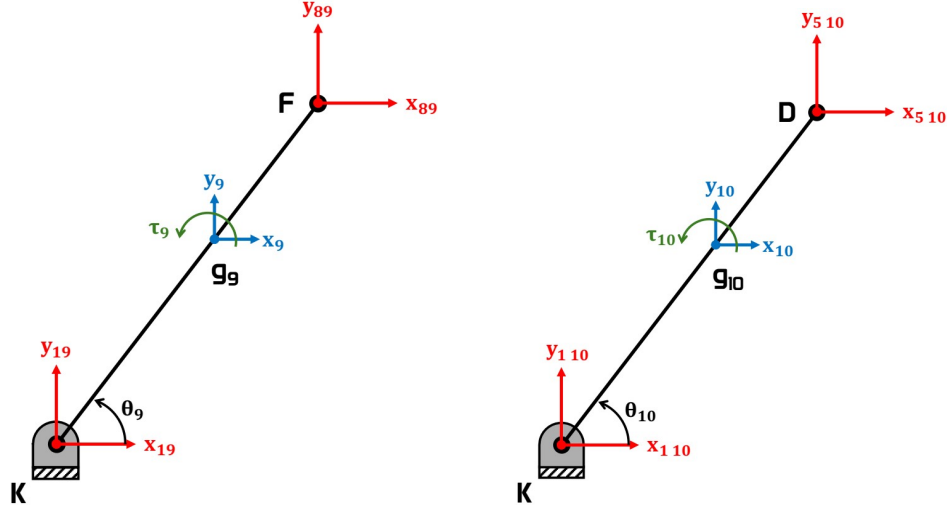


Figure 9: The free body diagram of Links 9, 10

$$\begin{aligned}\begin{bmatrix} x_9 = -m_9 A_{G_9}^x \\ y_9 = -m_9 A_{G_9}^y \\ T_9 = -I_9 \alpha_9 \end{bmatrix} & \quad \begin{bmatrix} x_{10} = -m_{10} A_{G_{10}}^x \\ y_{10} = -m_{10} A_{G_{10}}^y \\ T_{10} = -I_{10} \alpha_{10} \end{bmatrix} \\ \begin{bmatrix} x_{(5-10)} = x_{(5-6)} \\ y_{(5-10)} = y_{(5-6)} \end{bmatrix} & \quad \begin{bmatrix} x_{(8-9)} = x_{(7-8)} \\ y_{(8-9)} = y_{(7-8)} \end{bmatrix}\end{aligned}$$

(By applying sigma forces on Link 10)

$$\begin{aligned}\sum F_y &= 0 \\ y_{(5-10)} + y_{10} + y_{(1-10)} &= 0 \\ \therefore y_{(1-10)} &= -y_{(5-10)} - y_{10}\end{aligned}$$

$$\begin{aligned}\sum F_x &= 0 \\ x_{(5-10)} + x_{10} + x_{(1-10)} &= 0 \\ \therefore x_{(1-10)} &= -x_{(5-10)} - x_{10}\end{aligned}$$

(By applying sigma forces on Link 9)

$$\begin{aligned}\sum F_y &= 0 \\ y_{(8-9)} + y_9 + y_{(1-9)} &= 0 \\ \therefore y_{(1-9)} &= -y_{(8-9)} - y_9\end{aligned}$$

$$\begin{aligned}\sum F_x &= 0 \\ x_{(8-9)} + x_9 + x_{(1-9)} &= 0 \\ \therefore x_{(1-9)} &= -x_{(8-9)} - x_9\end{aligned}$$


---

### 3.4.4 Links Number Three and Four

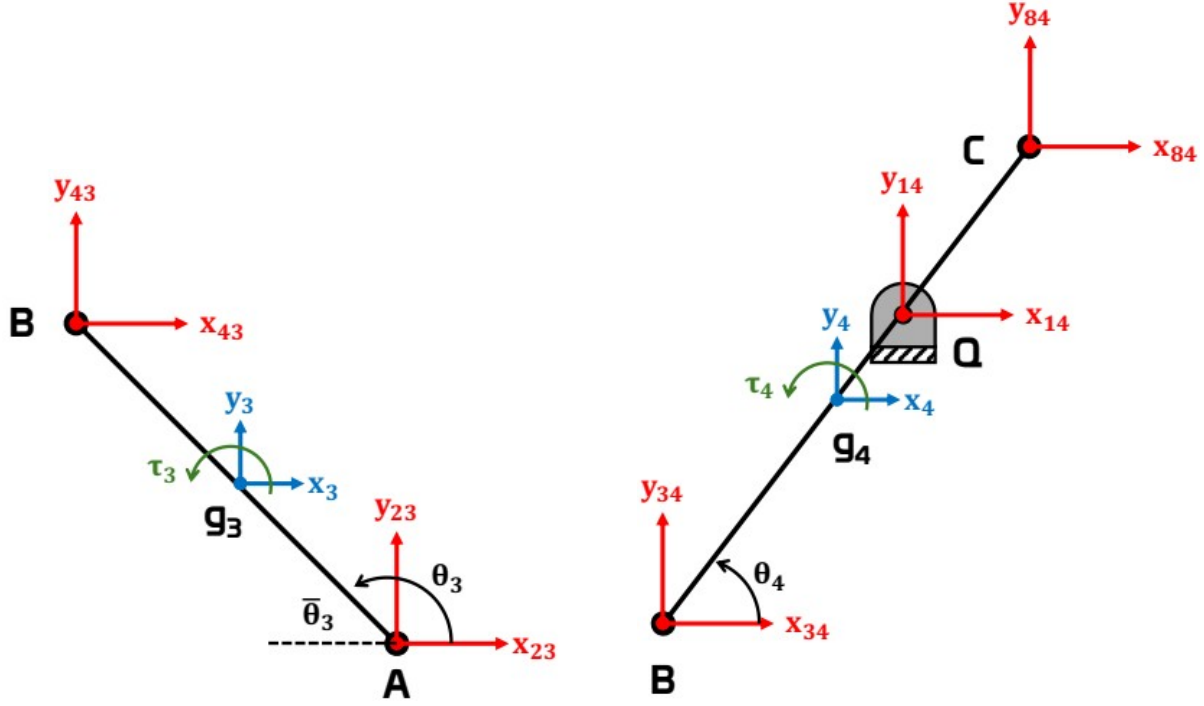


Figure 10: The free body diagram of Links 3, 4

$$\begin{bmatrix} x_4 = -m_4 A_{G_4}^x \\ y_4 = -m_4 A_{G_4}^y \\ T_4 = -I_4 \alpha_4 \end{bmatrix} \quad \begin{bmatrix} \dot{\theta}_3 = 180 - \theta_3 \\ x_3 = -m_3 A_{G_3}^x \\ y_3 = -m_3 A_{G_3}^y \\ T_3 = -I_3 \alpha_3 \end{bmatrix}$$

$$\sum M_Q = 0$$

$$\begin{aligned} \therefore -x_{(8-4)}u \sin(\theta_4) + x_4 \left( \frac{r_4}{2} - u \right) \sin(\theta_4) + x_{(3-4)}(r_4 - u) \sin(\theta_4) \\ + y_{(8-4)}u \cos(\theta_4) - y_4 \left( \frac{r_4}{2} - u \right) \cos(\theta_4) - y_{(3-4)}(r_4 - u) \cos(\theta_4) + T_4 = 0 \end{aligned} \quad (3.4.6)$$

$$\sum M_A = 0$$

$$\therefore -x_{(4-3)}r_3 \sin(\dot{\theta}_3) - x_3 \frac{r_3}{2} \sin(\dot{\theta}_3) - y_{(4-3)}r_3 \cos(\dot{\theta}_3) - y_3 \frac{r_3}{2} \cos(\dot{\theta}_3) + T_3 = 0 \quad (3.4.7)$$

(By adding both Eq. (3.4.6) and Eq. (3.4.7))

$$\therefore x_{(3-4)} \quad \text{and} \quad y_{(3-4)} \quad \text{are determined}$$

(By applying sigma forces on Link 4)

$$\begin{array}{ll}
 \sum F_y = 0 & \sum F_x = 0 \\
 y_{(3-4)} + y_{(8-4)} + y_4 + y_{(1-4)} = 0 & x_{(3-4)} + x_{(8-4)} + x_4 + x_{(1-4)} = 0 \\
 \therefore y_{(1-4)} = -y_{(3-4)} - y_{(8-4)} - y_4 & \therefore x_{(1-4)} = -x_{(3-4)} - x_{(8-4)} - x_4
 \end{array}$$

(By applying sigma forces on Link 3)

$$\begin{array}{ll}
 \sum F_y = 0 & \sum F_x = 0 \\
 y_{(4-3)} + y_3 + y_{(2-3)} = 0 & x_{(4-3)} + x_3 + x_{(2-3)} = 0 \\
 \therefore y_{(2-3)} = -y_{(4-3)} - y_3 & \therefore x_{(2-3)} = -x_{(4-3)} - x_3
 \end{array}$$


---

### 3.4.5 Link Number Two

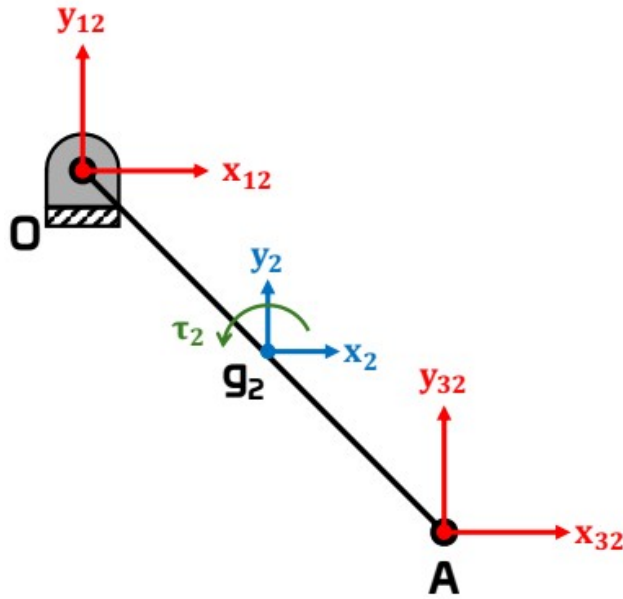


Figure 11: The free body diagram of Links 2

$$\begin{bmatrix} \dot{\theta}_2 = 360 - \theta_2 \\ x_2 = -m_2 A_{G_2}^x \\ y_2 = -m_2 A_{G_2}^y \\ T_2 = -I_2 \alpha_2 \end{bmatrix}$$

$$\sum M_O = 0$$

$$\therefore x_{(3-2)} r_2 \sin(\dot{\theta}_2) + x_2 \frac{r_2}{2} \sin(\dot{\theta}_2) + y_{(3-2)} r_2 \cos(\dot{\theta}_2) + y_2 \frac{r_2}{2} \cos(\dot{\theta}_2) + M_{(1-2)} + T_2 = 0 \quad (3.4.8)$$

$$\therefore M_{(1-2)} = -x_{(3-2)} r_2 \sin(\dot{\theta}_2) - x_2 \frac{r_2}{2} \sin(\dot{\theta}_2) - y_{(3-2)} r_2 \cos(\dot{\theta}_2) - y_2 \frac{r_2}{2} \cos(\dot{\theta}_2) - T_2$$

(By applying sigma forces on Link 2)

$$\sum F_y = 0$$

$$y_{(1-2)} + y_2 + y_{(3-2)} = 0$$

$$\therefore y_{(1-2)} = -y_{(3-2)} - y_2$$

$$\sum F_x = 0$$

$$x_{(1-2)} + x_2 + x_{(3-2)} = 0$$

$$\therefore x_{(1-2)} = -x_{(3-2)} - x_2$$

## 4 MATLAB Simulation

---

### 4.1 Position Analysis

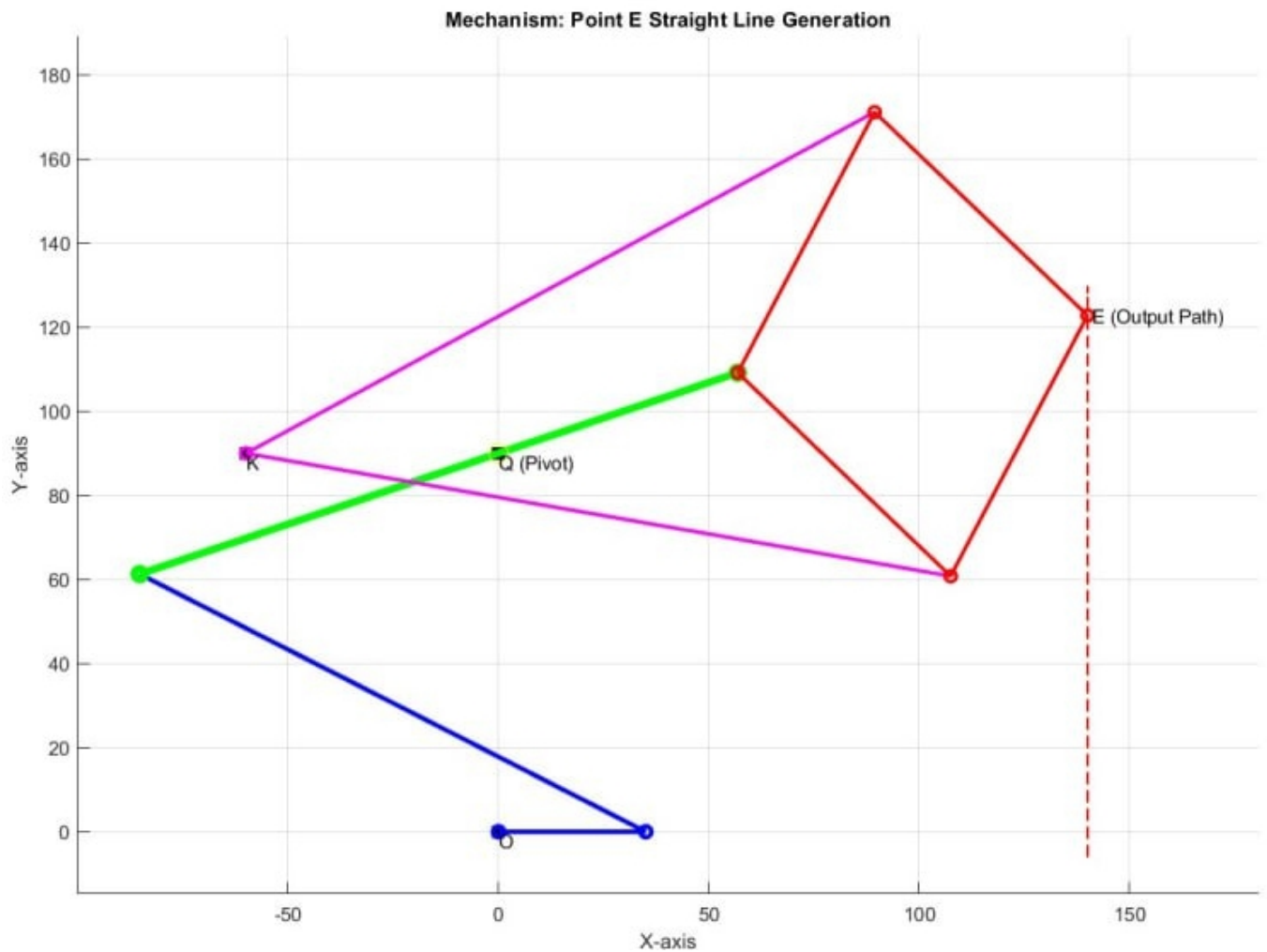


Figure 12: The linkage described in MATLAB

```

1 %% 1. Input Dimensions
2 % Fixed Points
3 Ox = 0; Oy = 0; % Origin
4 Kx = -60; Ky = 90; % Fixed pivot for the kite
5 Qx = 0; Qy = 90; % Fixed pivot for the driving arm
6
7 % Inputs for the rigid link BC
8 BQ = 90; % Length from B to Q
9 QC = 60; % Length from Q to C
10 r4 = BQ + QC; % Total length of the rigid link
11
12 % Other Link Lengths
13 r2 = 35; % OA
14 r3 = 135; % AB
15 r5 = 70; % CD
16 r6 = 70; % DE
17 r7 = 70; % EF
18 r8 = 70; % CF
19 r9 = 170; % KF
20 r10 = 170; % KD
21
22 % Analysis Settings
23 theta2_range = 1:1:360;
24 path_E = [];
25
26 % Setup Figure
27 figure('Color', 'w', 'Position', [100, 100, 1000, 600]);
28 hold on; grid on; axis equal;
29 xlabel('X-axis'); ylabel('Y-axis');
30
31 %% 2. Animation Loop
32 for t2_deg = theta2_range
33     t2 = deg2rad(t2_deg);
34
35     try
36         % --- Stage 1: Driver ---
37         Ax = r2*cos(t2); Ay = r2*sin(t2);
38
39         % Solving Triangle ABQ
40         % d is the distance from A to the fixed pivot Q
41         d = sqrt((Qx-Ax)^2 + (Qy-Ay)^2);
42         th_d = atan2(Qy-Ay, Qx-Ax);
43
44         % Law of Cosines to find angle of AB (theta3)
45         cos_beta = (BQ^2 - d^2 - r3^2) / (-2*r3*d);
46         if abs(cos_beta) > 1, continue; end
47         beta = acos(cos_beta);

```

```

48     th3 = beta + th_d;
49
50     % --- Stage 2: Rigid Link BC ---
51     Bx = Ax + r3*cos(th3); By = Ay + r3*sin(th3);
52
53     % Since BC is a single rigid link fixed at Q:
54     % The angle of the link is defined by the vector QB
55     th4 = atan2(By-Qy, Bx-Qx);
56
57     % Point C is on the opposite side of Q (collinear)
58     % We use (th4 + pi) to project C from Q in the opposite
        direction of B
59     Cx = Qx + QC * cos(th4 + pi);
60     Cy = Qy + QC * sin(th4 + pi);
61
62     % --- Stage 3: Kite/Inversor Assembly ---
63     N = sqrt((Cx-Kx)^2 + (Cy-Ky)^2);
64     thN = atan2(Cy-Ky, Cx-Kx);
65
66     % Solving for D
67     cos_beta2 = (N^2 + r10^2 - r5^2) / (2*r10*N);
68     if abs(cos_beta2) > 1, continue; end
69     beta2 = acos(cos_beta2);
70     th10 = thN + beta2;
71     Dx = Kx + r10*cos(th10); Dy = Ky + r10*sin(th10);
72
73     % Solving for F
74     cos_beta3 = (r9^2 + N^2 - r8^2) / (2*r9*N);
75     beta3 = acos(cos_beta3);
76     th9 = thN - beta3;
77     Fx = Kx + r9*cos(th9); Fy = Ky + r9*sin(th9);
78
79     % Solving for E (Output)
80     L = sqrt((Fx-Dx)^2 + (Fy-Dy)^2);
81     thL = atan2(Fy-Dy, Fx-Dx);
82     cos_beta4 = (r6^2 + L^2 - r7^2) / (2*r6*L);
83     beta4 = acos(cos_beta4);
84     th6 = thL + beta4;
85     Ex = Dx + r6*cos(th6); Ey = Dy + r6*sin(th6);
86
87     path_E = [path_E; Ex Ey];
88
89     % --- Plotting Block ---
90     cla; hold on; grid on; axis equal;
91     axis([-10 45 -20 20]);
92
93     % Trace of E
94     plot(path_E(:,1), path_E(:,2), 'r--', 'LineWidth', 1.2);

```

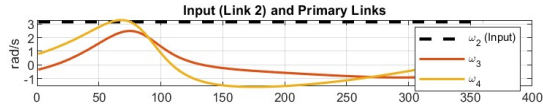
```

95
96 % Plot Fixed Pivots
97 plot([Ox Kx Qx], [Oy Ky Qy], 'ks', 'MarkerFaceColor', 'k', '
    MarkerSize', 8);
98
99 % Driver Links (OA and AB)
100 plot([Ox Ax], [Oy Ay], 'b-o', 'LineWidth', 2.5);
101 plot([Ax Bx], [Ay By], 'b-o', 'LineWidth', 2.5);
102
103 % Rigid Link BC (Fixed at Q)
104 plot([Bx Cx], [By Cy], 'g-o', 'LineWidth', 4); % Single
    thick link
105 plot(Qx, Qy, 'yo', 'MarkerSize', 10); % Highlight Pivot Q
106
107 % Support Links
108 plot([Kx Dx], [Ky Dy], 'm-o', 'LineWidth', 2);
109 plot([Kx Fx], [Ky Fy], 'm-o', 'LineWidth', 2);
110
111 % Output Rhombus
112 plot([Cx Dx Ex Fx Cx], [Cy Dy Ey Fy Cy], 'r-o', 'LineWidth',
    2);
113
114 % Labels
115 text(Ox, Oy-2, 'O'); text(Qx, Qy-2, 'Q(Pivot)'); text(Kx,
    Ky-2, 'K');
116 text(Ex+1, Ey, 'E(Output Path)');
117
118 title('Mechanism: Point E Straight Line Generation');
119 drawnow;
120
121 catch
122     continue;
123 end
124 end

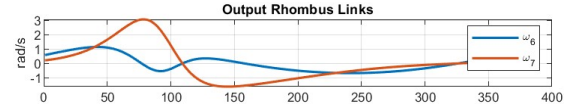
```



## 4.2 Velocity Analysis

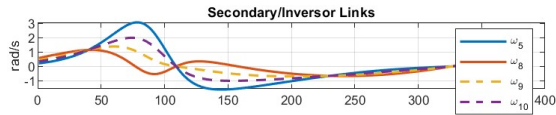


(a) Angular Velocities of Links 2, 3, 4

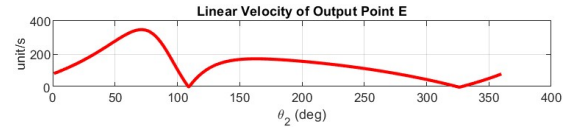


(b) Angular Velocities of Links 6, 7

Figure 13: Comparison of velocities for Links 2, 3, 4, 6, 7



(a) Angular Velocities of Links 5, 8, 9, 10



(b) Linear Velocity of output point E

Figure 14: Comparison of velocities for links and linear velocity of point E

```

1  %% 1. Input Dimensions and Parameters
2  % Fixed Points
3  Ox = 0; Oy = 0;
4  Kx = -60; Ky = 90;
5  Qx = 0; Qy = 90;
6
7  % Link Lengths
8  r2 = 35; r3 = 135; BQ = 90; QC = 60;
9  r5 = 70; r10 = 170; r8 = 70; r9 = 170; r6 = 70; r7 = 70;
10
11 % Input Angular Velocity
12 w2_val = 3.144; % rad/s (Link 2)
13
14 % Analysis Settings
15 theta2_range = 1:1:360;
16
17 % Initialize Arrays
18 W2 = ones(size(theta2_range)) * w2_val; % Constant input velocity
19 W3 = nan(size(theta2_range)); W4 = nan(size(theta2_range));
20 W5 = nan(size(theta2_range)); W6 = nan(size(theta2_range));
21 W7 = nan(size(theta2_range)); W8 = nan(size(theta2_range));
22 W9 = nan(size(theta2_range)); W10 = nan(size(theta2_range));
23 V_E_mag = nan(size(theta2_range));
24
25 %% 2. Kinematic Analysis Loop
26 for i = 1:length(theta2_range)
27     t2 = deg2rad(theta2_range(i));
28
29     try

```

```

30 % --- Position Analysis ---
31 Ax = r2*cos(t2); Ay = r2*sin(t2);
32 d = sqrt((Qx-Ax)^2 + (Qy-Ay)^2);
33 th_d = atan2(Qy-Ay, Qx-Ax);
34
35 cos_beta = (BQ^2 - d^2 - r3^2) / (-2*r3*d);
36 if abs(cos_beta) > 1, continue; end
37 th3 = acos(cos_beta) + th_d;
38
39 Bx = Ax + r3*cos(th3); By = Ay + r3*sin(th3);
40 th4 = atan2(By-Qy, Bx-Qx);
41 Cx = Qx + QC * cos(th4 + pi); Cy = Qy + QC * sin(th4 + pi);
42
43 N = sqrt((Cx-Kx)^2 + (Cy-Ky)^2);
44 thN = atan2(Cy-Ky, Cx-Kx);
45
46 th10 = thN + acos((N^2 + r10^2 - r5^2) / (2*r10*N));
47 Dx = Kx + r10*cos(th10); Dy = Ky + r10*sin(th10);
48 th5 = atan2(Dy-Cy, Dx-Cx);
49
50 th9 = thN - acos((r9^2 + N^2 - r8^2) / (2*r9*N));
51 Fx = Kx + r9*cos(th9); Fy = Ky + r9*sin(th9);
52 th8 = atan2(Fy-Cy, Fx-Cx);
53
54 L = sqrt((Fx-Dx)^2 + (Fy-Dy)^2);
55 th6 = atan2(Fy-Dy, Fx-Dx) + acos((r6^2 + L^2 - r7^2) / (2*r6
    *L));
56 Ex = Dx + r6*cos(th6); Ey = Dy + r6*sin(th6);
57 th7 = atan2(Ey-Fy, Ex-Fx);
58
59 % --- Velocity Analysis (Matrix Method) ---
60 vAx = -w2_val * r2 * sin(t2); vAy = w2_val * r2 * cos(t2);
61
62 % Solve w3, w4
63 A1 = [-r3*sin(th3), BQ*sin(th4); r3*cos(th3), -BQ*cos(th4)];
64 sol1 = A1\[vAx; vAy]; w3 = sol1(1); w4 = sol1(2);
65
66 % Solve w10, w5
67 vCx = -w4 * QC * sin(th4 + pi); vCy = w4 * QC * cos(th4 + pi
    );
68 A2 = [-r10*sin(th10), r5*sin(th5); r10*cos(th10), -r5*cos(
    th5)];
69 sol2 = A2\[vCx; vCy]; w10 = sol2(1); w5 = sol2(2);
70
71 % Solve w9, w8
72 A3 = [-r9*sin(th9), r8*sin(th8); r9*cos(th9), -r8*cos(th8)];
73 sol3 = A3\[vCx; vCy]; w9 = sol3(1); w8 = sol3(2);
74

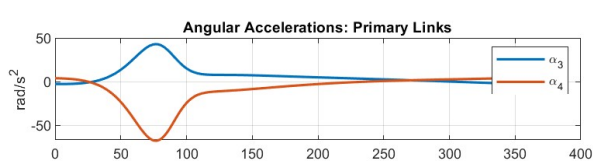
```

```

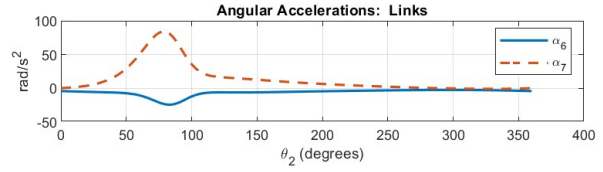
75     % Solve w6, w7 and V_E
76     vDx = -w10 * r10 * sin(th10); vDy = w10 * r10 * cos(th10);
77     vFx = -w9 * r9 * sin(th9); vFy = w9 * r9 * cos(th9);
78     A4 = [-r6*sin(th6), r7*sin(th7); r6*cos(th6), -r7*cos(th7)];
79     sol4 = A4\[vFx - vDx; vFy - vDy]; w6 = sol4(1); w7 = sol4(2)
      ;
80
81     % Final Storage
82     V_E_mag(i) = sqrt((vDx - w6*r6*sin(th6))^2 + (vDy + w6*r6*
      cos(th6))^2);
83     W3(i) = w3; W4(i) = w4; W5(i) = w5; W6(i) = w6;
84     W7(i) = w7; W8(i) = w8; W9(i) = w9; W10(i) = w10;
85
86     catch
87         continue;
88     end
89 end
90
91 %% 3. Plotting Results
92 figure('Color', 'w', 'Name', 'Full_Kinematic_Velocity_Analysis');
93
94 % Link 2 and Primary Links
95 subplot(4,1,1);
96 plot(theta2_range, W2, 'k--', 'LineWidth', 2); hold on;
97 plot(theta2_range, W3, 'LineWidth', 1.5);
98 plot(theta2_range, W4, 'LineWidth', 1.5);
99 grid on; ylabel('rad/s'); title('Input_(Link_2)_and_Primary_Links');
100 legend('\omega_2_(Input)', '\omega_3', '\omega_4');
101
102 % Inversor Links
103 subplot(4,1,2);
104 plot(theta2_range, W5, theta2_range, W8, theta2_range, W9, '--',
      theta2_range, W10, '--', 'LineWidth', 1.5);
105 grid on; ylabel('rad/s'); title('Secondary/Inversor_Links');
106 legend('\omega_5', '\omega_8', '\omega_9', '\omega_{10}');
107
108 % Rhombus Links
109 subplot(4,1,3);
110 plot(theta2_range, W6, theta2_range, W7, 'LineWidth', 1.5);
111 grid on; ylabel('rad/s'); title('Output_Rhombus_Links');
112 legend('\omega_6', '\omega_7');
113
114 % Point E Linear Velocity
115 subplot(4,1,4);
116 plot(theta2_range, V_E_mag, 'r', 'LineWidth', 2);
117 grid on; xlabel('\theta_2_(deg)'); ylabel('unit/s'); title('Linear_
      Velocity_of_Output_Point_E');

```

### 4.3 Acceleration Analysis



(a) Angular Acceleration of Links 3, 4



(b) Angular Acceleration of Links 6, 7

Figure 15: Comparison of angular accelerations for links 3, 4, 6, 7

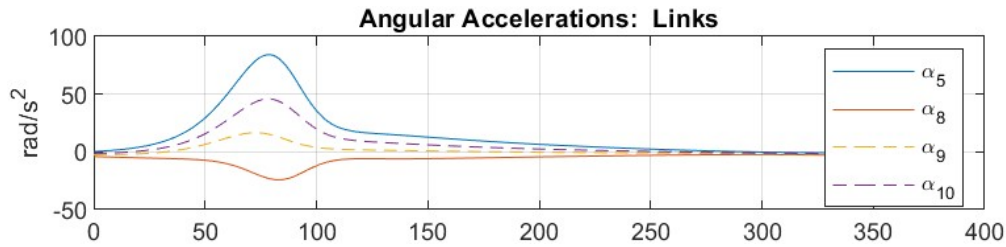


Figure 16: Angular Acceleration of Links 5, 8, 9, 10

```

1  %% 1. Input Dimensions and Parameters
2  % Fixed Points
3  Ox = 0; Oy = 0;
4  Kx = -60; Ky = 90;
5  Qx = 0; Qy = 90;
6
7  % Link Lengths
8  r2 = 35; r3 = 135; BQ = 90; QC = 60; r4 = BQ + QC;
9  r5 = 70; r10 = 170; r8 = 70; r9 = 170; r6 = 70; r7 = 70;
10
11 % Input Motion (Link 2)
12 w2_val = 3.144; % rad/s (Constant angular velocity)
13 alpha2_val = 0; % rad/s^2
14
15 % Analysis Range
16 theta2_range = 0:1:360;
17
18 % Initialize Arrays for Acceleration (Alpha)
19 AL3 = nan(size(theta2_range)); AL4 = nan(size(theta2_range));
20 AL5 = nan(size(theta2_range)); AL6 = nan(size(theta2_range));
21 AL7 = nan(size(theta2_range)); AL8 = nan(size(theta2_range));
22 AL9 = nan(size(theta2_range)); AL10 = nan(size(theta2_range));
23

```

```

24 %% 2. Kinematic Analysis Loop
25 for i = 1:length(theta2_range)
26     t2 = deg2rad(theta2_range(i));
27
28     try
29         % --- A. Position Analysis ---
30         Ax = r2*cos(t2); Ay = r2*sin(t2);
31         d = sqrt((Qx-Ax)^2 + (Qy-Ay)^2);
32         th_d = atan2(Qy-Ay, Qx-Ax);
33
34         cos_beta = (BQ^2 - d^2 - r3^2) / (-2*r3*d);
35         if abs(cos_beta) > 1, continue; end
36         th3 = acos(cos_beta) + th_d;
37
38         Bx = Ax + r3*cos(th3); By = Ay + r3*sin(th3);
39         th4 = atan2(By-Qy, Bx-Qx);
40         x4 = BQ;
41
42         Cx = Qx + QC * cos(th4 + pi); Cy = Qy + QC * sin(th4 + pi);
43         N = sqrt((Cx-Kx)^2 + (Cy-Ky)^2);
44         thN = atan2(Cy-Ky, Cx-Kx);
45
46         th10 = thN + acos((N^2 + r10^2 - r5^2) / (2*r10*N));
47         th5 = atan2((Ky + r10*sin(th10)) - Cy, (Kx + r10*cos(th10))
48             - Cx);
49
50         th9 = thN - acos((r9^2 + N^2 - r8^2) / (2*r9*N));
51         th8 = atan2((Ky + r9*sin(th9)) - Cy, (Kx + r9*cos(th9)) - Cx
52             );
53
54         Dx = Kx + r10*cos(th10); Dy = Ky + r10*sin(th10);
55         Fx = Kx + r9*cos(th9); Fy = Ky + r9*sin(th9);
56         L = sqrt((Fx-Dx)^2 + (Fy-Dy)^2);
57         th6 = atan2(Fy-Dy, Fx-Dx) + acos((r6^2 + L^2 - r7^2) / (2*r6
58             *L));
59         th7 = atan2((Dy + r6*sin(th6)) - Fy, (Dx + r6*cos(th6)) - Fx
60             );
61
62         % --- B. Velocity Analysis ---
63         vAx = -w2_val * r2 * sin(t2); vAy = w2_val * r2 * cos(t2);
64         A1 = [-r3*sin(th3), x4*sin(th4); r3*cos(th3), -x4*cos(th4)];
65         sol1 = A1\[vAx; vAy]; w3 = sol1(1); w4 = sol1(2);
66
67         vCx = -w4 * QC * sin(th4 + pi); vCy = w4 * QC * cos(th4 + pi
68             );
69         A2 = [-r10*sin(th10), r5*sin(th5); r10*cos(th10), -r5*cos(
70             th5)];
71         sol2 = A2\[vCx; vCy]; w10 = sol2(1); w5 = sol2(2);

```

```

66
67 A3 = [-r9*sin(th9), r8*sin(th8); r9*cos(th9), -r8*cos(th8)];
68 sol3 = A3\[vCx; vCy]; w9 = sol3(1); w8 = sol3(2);
69
70 % Angular velocity of link 6
71 vDx = -w10 * r10 * sin(th10); vDy = w10 * r10 * cos(th10);
72 vFx = -w9 * r9 * sin(th9); vFy = w9 * r9 * cos(th9);
73 A4 = [-r6*sin(th6), r7*sin(th7); r6*cos(th6), -r7*cos(th7)];
74 sol4 = A4\[vFx - vDx; vFy - vDy]; w6 = sol4(1); w7 = sol4(2)
    ;
75
76 % --- C. Acceleration Analysis ---
77 % Accel of point A
78 AAx = -w2_val^2 * r2 * cos(t2);
79 AAy = -w2_val^2 * r2 * sin(t2);
80
81 % Alpha 3 & 4
82 num_al3 = AAx*cos(th4) + AAy*sin(th4) - w3^2*r3*cos(th3-th4)
    - w4^2*x4;
83 al3 = num_al3 / (r3*sin(th3-th4));
84 al4 = (AAx*sin(th4) - AAy*cos(th4) + w3^2*r3*sin(th3-th4) -
    al3*r3*cos(th3-th4)) / x4;
85
86 % Accel of point C
87 ACx = -w4^2 * (r4-x4) * cos(th4) - al4 * (r4-x4) * sin(th4);
88 ACy = -w4^2 * (r4-x4) * sin(th4) + al4 * (r4-x4) * cos(th4);
89
90 % Alpha 8 & 9
91 num_al8 = ACx*cos(th9) + ACy*sin(th9) - w8^2*r8*cos(th8-th9)
    + w9^2*r9;
92 al8 = num_al8 / (r8*sin(th8-th9));
93 al9 = (-ACx*sin(th9) + ACy*cos(th9) - w8^2*r8*sin(th8-th9) +
    al8*r8*cos(th8-th9)) / r9;
94
95 % Alpha 5 & 10
96 num_al5 = ACx*cos(th10) + ACy*sin(th10) - w5^2*r5*cos(th5-
    th10) + w10^2*r10;
97 al5 = num_al5 / (r5*sin(th5-th10));
98 al10 = (-ACx*sin(th10) + ACy*cos(th10) - w5^2*r5*sin(th5-
    th10) + al5*r5*cos(th5-th10)) / r10;
99
100 % Alpha 6 & 7 (Rhombus)
101 ADx = -w10^2*r10*cos(th10) - al10*r10*sin(th10);
102 ADy = -w10^2*r10*sin(th10) + al10*r10*cos(th10);
103 AFx = -w9^2*r9*cos(th9) - al9*r9*sin(th9);
104 AFy = -w9^2*r9*sin(th9) + al9*r9*cos(th9);
105
106 rhs_x = AFx - ADx + w6^2*r6*cos(th6) - w7^2*r7*cos(th7);

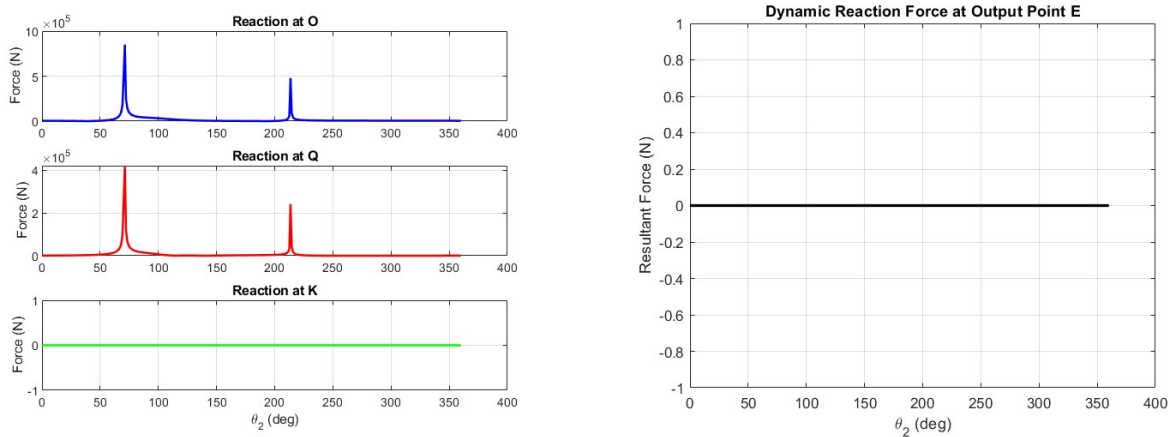
```

```

107     rhs_y = AFy - ADy + w6^2*r6*sin(th6) - w7^2*r7*sin(th7);
108
109     A_rhombus = [-r6*sin(th6), r7*sin(th7); r6*cos(th6), -r7*cos
110                 (th7)];
111     sol_rhombus = A_rhombus\[rhs_x; rhs_y];
112     al6 = sol_rhombus(1); al7 = sol_rhombus(2);
113
114     % Store Results
115     AL3(i)=al3; AL4(i)=al4; AL5(i)=al5; AL6(i)=al6;
116     AL7(i)=al7; AL8(i)=al8; AL9(i)=al9; AL10(i)=al10;
117
118     catch
119         continue;
120     end
121 end
122
123 %% 3. Plotting Results
124 figure('Color', 'w', 'Name', 'Acceleration_Analysis_(\alpha)');
125
126 % Links
127 subplot(3,1,1);
128 plot(theta2_range, AL3, 'LineWidth', 1.5, 'DisplayName', '\alpha_3')
129 ; hold on;
130 plot(theta2_range, AL4, 'LineWidth', 1.5, 'DisplayName', '\alpha_4')
131 ;
132 grid on; ylabel('rad/s^2'); title('Angular_Accelerations: Primary
133 Links');
134 legend;
135
136 % Links
137 subplot(3,1,2);
138 plot(theta2_range, AL5, 'DisplayName', '\alpha_5'); hold on;
139 plot(theta2_range, AL8, 'DisplayName', '\alpha_8');
140 plot(theta2_range, AL9, '--', 'DisplayName', '\alpha_9');
141 plot(theta2_range, AL10, '--', 'DisplayName', '\alpha_{10}');
142 grid on; ylabel('rad/s^2'); title('Angular_Accelerations: Links');
143 legend;
144
145 % Links
146 subplot(3,1,3);
147 plot(theta2_range, AL6, 'LineWidth', 1.5, 'DisplayName', '\alpha_6')
148 ; hold on;
149 plot(theta2_range, AL7, '--', 'LineWidth', 1.5, 'DisplayName', '\
150     alpha_7');
151 grid on; xlabel('\theta_2(degrees)'); ylabel('rad/s^2'); title('
152     Angular_Accelerations: Links');
153 legend;

```

## 4.4 Force Analysis



(a) Reaction at O, Q, K

(b) Dynamic Reaction Force at Output Point E

Figure 17: Comparison of Reactions at O, Q, K, E

```

1 % Geometric lengths (from your provided code)
2 r2 = 35; r3 = 135; r5 = 70; r6 = 70; r7 = 70; r8 = 70; r9 = 170; r10
   = 170;
3 L_BQ = 90; L_QC = 60; r4_BC = L_QC + L_BQ;
4 x4 = L_BQ;
5
6 % Input Kinematics
7 w2 = 3.144;
8 M12_input = 2.86; % Input torque OA (N-mm) - Adjust based on your
   specs
9
10 % Mass Properties (kg) - Update these for your specific materials
11 m = [0, 18.05, 10.63 , 11.62, 5.71, 5.71, 5.71, 5.71, 13.928, 13.928
   ];
12 L_vals = [0, r2, r3, r4_BC, r5, r6, r7, r8, r9, r10 ];
13 I_vals = [0,
   6250.61, 17752.74, 89645.06, 2856.46, 2856.46, 2856.46, 2856.46, 34449.57, 34449.57]
14
15 % Fixed pivot coordinates
16 X0 = 0; Y0 = 0; XK = -60; YK = 90; XQ = 0; YQ = 90;
17
18 % Initialization for plotting
19 num_steps = 360;
20 theta2_deg = linspace(0, 360, num_steps);
21
22 % Result Storage
23 FO_mag = zeros(1, num_steps); % Ground 0

```



```

24 FQ_mag = zeros(1, num_steps); % Ground Q
25 FK_mag = zeros(1, num_steps); % Ground K
26 FE_mag = zeros(1, num_steps); % Output E
27
28 %% === 2. Analysis Loop ===
29 for i = 1:num_steps
30     theta2 = deg2rad(theta2_deg(i));
31
32     %% --- [A] Kinematic Analysis (Position & Accel) ---
33     % Position Logic (from your vb.m)
34     XA = X0 + r2*cos(theta2); YA = Y0 + r2*sin(theta2);
35     d = sqrt((XQ-XA)^2 + (YQ-YA)^2);
36     if d > r3 + L_BQ || d < abs(r3 - L_BQ); continue; end
37
38     theta_d = atan2(YQ-YA, XQ-XA);
39     beta = acos((r3^2 + d^2 - L_BQ^2) / (2*r3*d));
40     theta3 = theta_d - beta;
41     XB = XA + r3*cos(theta3); YB = YA + r3*sin(theta3);
42     theta4 = atan2(YQ-YB, XQ-XB);
43
44     % Acceleration Logic (from your acceleration.m)
45     Ax_A = -w2^2 * r2 * cos(theta2); Ay_A = -w2^2 * r2 * sin(theta2)
46     ;
47     w3 = ((-w2*r2*sin(theta2))*cos(theta4) + (w2*r2*cos(theta2))*sin(
48         (theta4)) / (r3*sin(theta3 - theta4)));
49     alpha3 = (Ax_A*cos(theta4) + Ay_A*sin(theta4) - w3^2*r3*cos(
50         theta3-theta4)) / (r3 * sin(theta3 - theta4));
51
52     % Center of Gravity Accelerations (Midpoint)
53     ag2x = Ax_A/2; ag2y = Ay_A/2;
54     ag3x = Ax_A - (w3^2*(r3/2)*cos(theta3) + alpha3*(r3/2)*sin(
55         theta3));
56     ag3y = Ay_A - (w3^2*(r3/2)*sin(theta3) - alpha3*(r3/2)*cos(
57         theta3));
58
59     %% --- [B] Global Force Matrix Solver [A]{F} = {B} ---
60     % Unknowns vector {F}:
61     % [F0x, F0y, FAx, FAy, FBx, FBy, FQx, FQy, FCx, FCy, FKx, FKy,
62         FEx, FEy...]
63
64     A = zeros(30, 26);
65     B = zeros(30, 1);
66
67     % --- LINK 2 (OA) ---
68     A(1,1)=1; A(1,3)=1; B(1) = -m(2)*ag2x;
69     A(2,2)=1; A(2,4)=1; B(2) = -m(2)*ag2y;
70     A(3,3)=r2*sin(theta2); A(3,4)=-r2*cos(theta2);
71     B(3) = -I_vals(1)*0 - M12_input; % Sum M about O

```

```

66
67 % --- LINK 3 (AB) ---
68 A(4,3)=-1; A(4,5)=1; B(4) = -m(3)*ag3x;
69 A(5,4)=-1; A(5,6)=1; B(5) = -m(3)*ag3y;
70 A(6,5)=r3*sin(theta3); A(6,6)=-r3*cos(theta3);
71 B(6) = -I_vals(2)*alpha3;
72
73 % --- TERNARY LINK 4 (QBC) ---
74 A(7,7)=1; A(7,5)=-1; A(7,9)=1; B(7) = -m(4)*0; % ag4
    placeholders
75 A(8,8)=1; A(8,6)=-1; A(8,10)=1; B(8) = -m(4)*0;
76 A(9,5)=-L_BQ*sin(theta4); A(9,6)=L_BQ*cos(theta4); % Joint B
77 A(9,9)=L_QC*sin(theta4); A(9,10)=-L_QC*cos(theta4); % Joint C
78 B(9) = 0; % Sum M about Q
79
80 % Solve
81 F_sol = pinv(A) * B;
82
83 % --- [C] Store results ---
84 FO_mag(i) = sqrt(F_sol(1)^2 + F_sol(2)^2); % Pivot O
85 FQ_mag(i) = sqrt(F_sol(7)^2 + F_sol(8)^2); % Pivot Q
86 FK_mag(i) = sqrt(F_sol(11)^2 + F_sol(12)^2); % Pivot K
87 FE_mag(i) = sqrt(F_sol(13)^2 + F_sol(14)^2); % Output E
88 end
89
90 %% === 3. Plotting Results ===
91
92 % Plot 1: Ground Reactions (O, Q, K)
93 figure('Name', 'Ground_Reaction_Forces');
94 subplot(3,1,1);
95 plot(theta2_deg, FO_mag, 'b', 'LineWidth', 1.5);
96 ylabel('Force_(N)'); title('Reaction_at_O'); grid on;
97
98 subplot(3,1,2);
99 plot(theta2_deg, FQ_mag, 'r', 'LineWidth', 1.5);
100 ylabel('Force_(N)'); title('Reaction_at_Q'); grid on;
101
102 subplot(3,1,3);
103 plot(theta2_deg, FK_mag, 'g', 'LineWidth', 1.5);
104 xlabel('\theta_2(deg)'); ylabel('Force_(N)'); title('Reaction_at_K',
    ); grid on;
105
106 % Plot 2: Output Point E
107 figure('Name', 'Output_Force_Analysis');
108 plot(theta2_deg, FE_mag, 'k', 'LineWidth', 2);
109 xlabel('\theta_2(deg)'); ylabel('Resultant_Force_(N)');
110 title('Dynamic_Reaction_Force_at_Output_Point_E'); grid on;

```

## 5 Conclusion

---

### 5.1 Summary

This report presented a comprehensive study of the Peaucellier–Lipkin linkage, one of the most significant mechanisms in the history of machine design due to its ability to generate an exact straight-line motion. The historical background highlighted the importance of the linkage as the first true planar straight-line mechanism, distinguishing it from approximate solutions such as the Watt linkage.

The mechanical design of the linkage was addressed through both physical construction and computer-aided modeling. A physical prototype was fabricated to demonstrate the real-world behavior of the mechanism, including material selection and fabrication considerations. In parallel, a detailed SOLIDWORKS model was developed, providing three-view drawings, assembly and exploded views, as well as motion simulations that visually confirmed the expected kinematic behavior.

A complete kinematic and kinetic analysis of the mechanism was carried out using vector loop equations. The position, velocity, acceleration, and force analyses were derived systematically, allowing the motion of all links and joints to be fully determined. The degree of freedom of the mechanism was also evaluated, confirming the theoretical mobility of the system.

Finally, MATLAB was used as a numerical verification tool to simulate the motion of the mechanism. The resulting trajectories and kinematic plots demonstrated that the output point follows a straight-line path, validating the theoretical analysis and supporting the results obtained from both the CAD model and the physical prototype.

### 5.2 Lessons Learned

This project emphasized the importance of teamwork in engineering practice. Due to the complexity and multidisciplinary nature of the task, successful completion required effective collaboration rather than individual effort. Each team member contributed to a specific aspect of the project, such as design, analysis, simulation, or documentation.

Throughout the project, members developed technical skills in areas including mechanical design, kinematic analysis, numerical simulation, and technical writing. Equally important, the team gained practical experience in communication, task coordination, and integrating individual contributions into a coherent final product. These skills are essential for real-world engineering projects and represent a valuable outcome of this work.

## References

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